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# Electromagnetic Waves

Sound is a compression wave — to travel, it needs a medium to compress: air, water, etc. (Regardless of what you have seen in movies, sound does not travel through a vacuum!)

Light is an electromagnetic wave — it causes fluctuations in the electric and magnetic fields that are everywhere. It can cross a vacuum, as it does to reach us from the sun.

Electromagnetic waves travel at about 300 million meters per second in a vacuum. The waves travel slower through different types of matter. For example, an electromagnetic wave travels at 225 million meters per second in water.

Electromagnetic waves come in different frequencies. For example, the light coming out of a red laser pointer is usually about  $4.75 \times 10^{14}$  Hz. The wifi data sent by your computer is carried on an electromagnetic wave too. It is usually close to  $2.4 \times 10^6$  Hz or  $5 \times 10^6$  Hz.

Because we know how fast the waves are moving, we sometimes talk about their wavelengths instead of their frequencies. Since we know its speed and its oscillation rate, we can calculate its wavelength. Since we want the distance per oscillation, we can divide the speed by the frequency. The light coming out of a laser pointer is  $300 \times 10^6 / 4.75 \times 10^{14} = 630 \times 10^{-9}$  m, or 630 nm.

### Exercise 1      Wavelengths

A green laser pointer emits light at  $5.66 \times 10^{14}$  Hz. What is its wavelength in a vacuum?

*Working Space*

*Answer on Page 75*

This can be simplified as:

$$\lambda = \frac{c}{f}$$

where  $c = 3 \times 10^8$  m/s (the speed of light in a vacuum)

We have given names to different ranges of the electromagnetic spectrum in Figure 1.1

| Name        | Hertz                   | Meters                  |
|-------------|-------------------------|-------------------------|
| Gamma rays  | $10^{20} - 10^{24}$     | $10^{-14} - 10^{-11}$   |
| X-rays      | $10^{16} - 10^{20}$     | $10^{-11} - 10^{-8}$    |
| Ultraviolet | $10^{15} - 10^{16}$     | $10^{-8} - 10^{-7}$     |
| Blue        | $\sim 6 \times 10^{14}$ | $\sim 5 \times 10^{-7}$ |
| Red         | $\sim 4 \times 10^{14}$ | $\sim 7 \times 10^{-7}$ |
| Infrared    | $10^{12} - 10^{14}$     | $10^{-6} - 10^{-3}$     |
| Microwaves  | $10^9 - 10^{12}$        | $10^{-3} - 10^{-1}$     |
| Radio waves | $10^3 - 10^9$           | $10^{-1} - 10^5$        |

Figure 1.1: A table of different ranges of the electromagnetic spectrum.

(You may have heard of “cosmic rays” and wonder why they are not listed in this table. Cosmic rays are actually the nuclei of atoms that have been stripped of their electron cloud. These particles come flying out of the sun at very high speeds. They were originally thought to be electromagnetic waves, and were mistakenly named “rays”.)

In general, the lower frequency the wave is, the better it passes through a mass. A radio wave, for example, can pass through the walls of your house, but visible light cannot. The people who designed the microwave oven chose the frequency of 2.45 GHz because the energy from those waves tended to get absorbed in the first few inches of food that it passed through.

## 1.1 The greenhouse effect

Humans have dug up a bunch of long carbon-based molecules (like oil and coal) and burned them, releasing large amounts of  $\text{CO}_2$  into the atmosphere. It may not be obvious why that has made the planet warmer, but the answer is electromagnetic waves.

A warm object gives off infrared electromagnetic waves. That’s why, for example, motion detectors in security systems are actually infrared detectors: even in a dark room, your body gives off a lot of infrared radiation.

You may have heard of “heat-seeking missiles.” These are more accurately called “Infrared homing missiles” because they follow objects giving off infrared radiation – hot things like jet engines.

The sun beams a lot of energy to our planet in the form of electromagnetic radiation: visible light, infrared, ultraviolet. (How much? At the top of the atmosphere directly facing the sun, we get 1,360 watts of radiation per square meter. That is a lot of power!)

Some of that radiation just reflects back into space. 23% is reflected by the clouds and the atmosphere, while 7% makes it all the way to the surface of the planet and is reflected back into space.

The other 71% is absorbed. 48% is absorbed by the surface and 23% is absorbed by the atmosphere. All of that energy warms the planet and the atmosphere so that it gives off infrared radiation. The planet lives in equilibrium; the infrared radiation leaving our atmosphere is exactly the same amount of energy as that 71% of the radiation that it absorbs.

(If the planet absorbs more energy than it releases, the planet gets hotter. Hotter things release more infrared. When the planet is in equilibrium again, it stops getting hotter.)

So, what is the problem with CO<sub>2</sub> and other large molecules in the atmosphere? They absorb the infrared radiation instead of letting it escape into space. This means the planet must be hotter to maintain equilibrium.

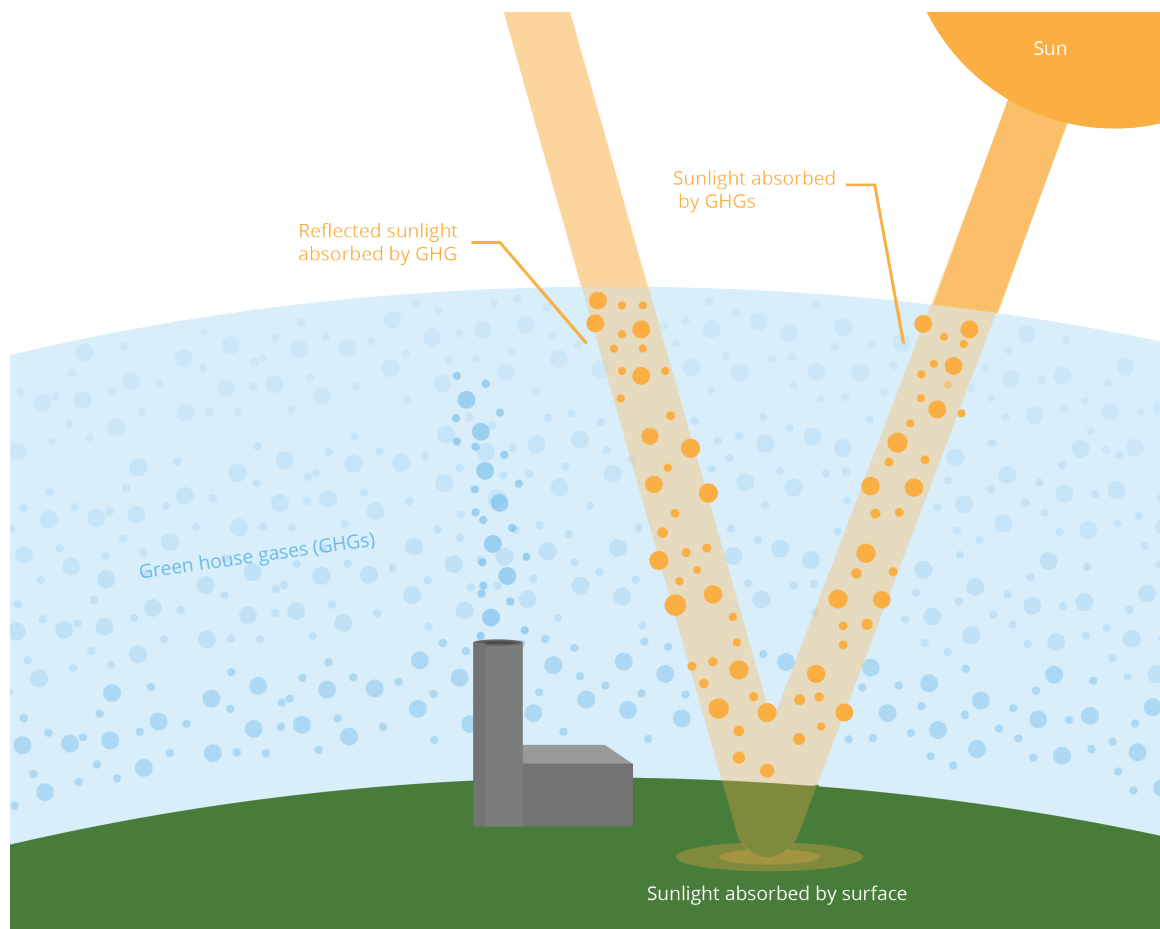


Figure 1.2: Greenhouse gas effects from the solar rays.

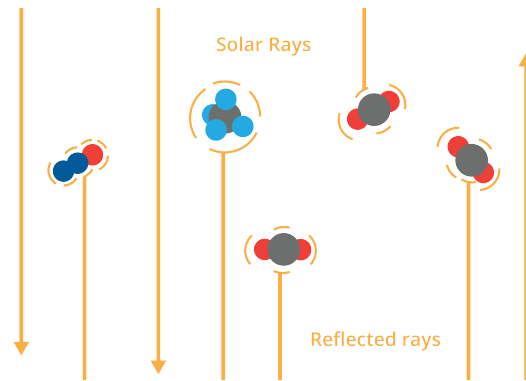


Figure 1.3: A zoomed in version of greenhouse gases.

The planet is getting hotter, and it is creating a multitude of problems:

- Weather patterns are changing, which leads to extreme floods and droughts.
- Ice and snow in places like Greenland are melting and flowing into the oceans. This is raising sea levels.
- Biomes with biodiversity are resilient. Rapidly changing climate is destroying biodiversity everywhere, which is making these ecosystems very fragile.
- In many places, permafrost, which has trapped large amounts of methane in the ground for millennia, is melting.

That last item is particularly scary, because methane is a large gas molecule — it absorbs even more infrared radiation than  $\text{CO}_2$ . As it escapes the permafrost, the problem will get worse.

Scientists are working on four kinds of solutions:

- **Stop increasing the amount of greenhouse gases in our atmosphere.** It is hoped that non-carbon based energy systems like solar, wind, hydroelectric, and nuclear could let us stop burning carbon. Given the methane already being released, it maybe too late for this solution to work on its own.
- **Take some of greenhouse gases out of our atmosphere and sequester them somewhere.** The trunk of a tree is largely made up of carbon molecules. When you grow a tree where there had not been one before, you are sequestering carbon inside the tree. There are also scientists that are trying to develop a process that pulls greenhouse gases out of the air and turn them into solids.
- **Decrease the amount of solar radiation that is absorbed by our planet and its atmosphere.** Clouds reflect a lot of radiation back into space. Could we increase the cloudiness of our atmosphere? Or maybe launch mirrors into orbit around our

planet?

- **Adapt to the changing climate.** These scientists are assuming that global warming will continue, and are working to minimize future human suffering. How will we relocate a billion people as the oceans claim their homes? When massive heat waves occur, how will we keep people from dying? As biodiversity decreases, how can we make sure that species that are important to human existence survive?

What are the greenhouse gases and how much does each contribute to keeping the heat from exiting to space? These numbers are still being debated, but this will give you a feel:

|                |                  |           |
|----------------|------------------|-----------|
| Water vapor    | H <sub>2</sub> O | 36 - 72 % |
| Carbon dioxide | CO <sub>2</sub>  | 9 - 26 %  |
| Methane        | CH <sub>4</sub>  | 4 - 9 %   |
| Ozone          | O <sub>3</sub>   | 3 - 7 %   |

Notice that while we talk a lot about carbon dioxide, the most important greenhouse gas is actually water. Why don't we talk about it? Given the enormous surfaces of the oceans, it is difficult to imagine any way to permanently decrease the amount of water in the air. Additionally, a great deal of water in the air is in the form of clouds, which help reflect radiation before it is absorbed.





# Reflection and Photons

What happens when light hits a mass?

## 2.1 Photons

In a previous chapter, we talked about light as a wave, and we mentioned that each color in the rainbow is a different wavelength. You can also think of light as particles of energy called *photons*. The term photons was coined by physicist Gilbert Newton Lewis, the same Lewis known for the Lewis dot structures we have talked about at the beginning of the sequence!

Every photon comes with an amount of energy that determines what color it is. When we are talking about light interacting with objects, your intuition will be right more often if you think of light as a beam of photons. These photons travel at the *vacuum speed of light*, more often called the *vacuum speed of light*. This speed, in empty space or vacuums, is given by  $2.9979 \times 10^8$  m/s. Photons also

- behave like particles and a wave.
- have zero mass
- has zero charge
- can be created and destroyed when radiation is present
- release electrons upon collisions.

### Formula for energy of a photon

For any given photon, its energy,  $E$ , is given by

$$E = hf = \frac{hc}{\lambda}$$

where  $c$  is the speed of light,  $\lambda$  is the wavelength of the electromagnetic wave, and  $h$  Planck's constant:  $6.63 \times 10^{-34}$  m<sup>2</sup>kg/s. Note that the angular frequency  $f = \frac{c}{\lambda}$ , and Planck's constant represents the quantum of action, establishing that energy is not continuous but exists in discrete packets called quantas. We will talk in a future

chapter about how charge and energy is quantized, or packeted in integerized values which we call discrete.

For example, a red laser light has a wavelength of about 630 nm. So, the energy in each photon is:

$$\frac{(300 \times 10^6)(6.63 \times 10^{-34})}{630 \times 10^{-9}} = 3.1 \times 10^{-19} \text{ joules}$$

For units sake, this is often abbreviated as  $E = \frac{1240}{\lambda}$  where  $\lambda$  is in nanometers and  $E$  is in electronvolts (eV).

### 2.1.1 The Photoelectric Effect

In classical physics, scientists like Planck and Einstein thought that if you shined a light on a metal long enough or brightened it enough, it would eventually knock an electron loose, like a wave slowly pushing a boat to shore. Experiments showed that if the light was the wrong color, meaning too low of a frequency, nothing happened, no matter how bright or powerful the light was.

This showed that light is quantized, and in order to travel up or down the energy levels of an atom, it must absorb or emit a photon with the exact energy difference between those levels. This phenomenon is known as the *photoelectric effect*.

The effect is governed by the conservation of energy. The energy of an incoming photon ( $E_{\text{photon}}$ ) must be greater than the bond holding the electron to the metal. This is described by

$$E_{\text{photon}} = \Phi + K_{\text{max}}$$

where

- $E_{\text{photon}} = hf$ : The energy of the incoming photon, where  $h$  is Planck's constant and  $f$  is the frequency.
- $\Phi$  (*Work Function*): The minimum energy required to eject an electron from a specific metal. Each material has a different "cost" to let an electron go.
- $f_0$  (*Threshold Frequency*): The specific frequency where the photon has exactly enough energy to match the work function ( $\Phi = hf_0$ ). If  $f < f_0$ , no electrons are ejected. If  $f = f_0$ , electrons are released but have zero kinetic energy ( $K_{\text{max}} = 0$ ).
- $K_{\text{max}}$ : The "leftover" kinetic energy of the electron once released from the metal.

This is how solar panels and digital camera sensors turn light into electricity. Solar panels work like a one-way road for light energy. First, sunlight hits the panel in the form of photons, which act like tiny kicks that knock electrons loose from their atoms. Because the panel has a built-in internal electric field, these loose electrons are forced to move in only one direction (back into the panel), creating a steady flow of Direct Current (DC) electricity that can power your home. We are going to look at camera sensors in depth in a future workbook.

### 2.1.2 Photons and Momentum

In classical mechanics, we define momentum as the product of mass and velocity ( $p = mv$ ).

This leads to a common question: if a photon has zero rest mass, how can it possess momentum? Energy and momentum are linked, even for massless particles. For a photon, the momentum ( $p$ ) is inversely proportional to its wavelength ( $\lambda$ ):

$$p = \frac{h}{\lambda} \quad \text{or} \quad p = \frac{E}{c} \quad (2.1)$$

**Application:** For example, solar sails are spacecraft propulsion systems in space. These large, mirror-like sails in space can intercept millions of photons. As these photons reflect off the surface, they undergo a change in momentum ( $\Delta p$ ), which, according to Newton's Second Law ( $F = \frac{\Delta p}{\Delta t}$ ), exerts a constant force on the spacecraft, allowing for fuel-free propulsion.

### Photonic Interactions

When a photon comes from the sun and hits an object, one of several things can happen:

- The energy of the photon is absorbed by the object. It makes the object a little warmer. If a large proportion of photons hitting the mass are absorbed like this, we say the object is “black”.
- The photon bounces off the object. If the surface is very smooth, the photons bounce in a predictable manner, and we call this *reflection* and we say the object is “shiny”.
- If the surface is rough and the photons are not absorbed, the photons are scattered in random directions. We call this *diffusion*. If most of the photons hitting an object are bounced in random directions, we say that the object is “white”.
- The photon passes through the mass. If the mass has smooth surfaces and a consistent composition, the photons will pass through the mass in a predictable manner. We say that the mass is “transparent”.

- If the photons pass through, but in an unpredictable, scattering manner, we say the mass is “translucent”.

No object absorbs every photon, but chemists are always coming up with “blackier” materials. Vantablack, for example, is a super-black paint that absorbs 99.965% of all photons in the visible spectrum.

No object reflects every photon, but a mirror is pretty close. Let’s talk about reflections in a mirror.

## 2.2 Reflection

When a beam of light hits a mirror, it bounces off the mirror at the same angle it approached from. That is, if it approaches nearly perpendicular to the mirror, it departs nearly perpendicular to the mirror. If it hits the mirror at a glancing angle, it departs at an angle close to the mirror’s surface. A good video to see this in action can be found here: [https://www.youtube.com/shorts/qA\\_VQZfTiUA](https://www.youtube.com/shorts/qA_VQZfTiUA)

### 2.2.1 Law of Reflection

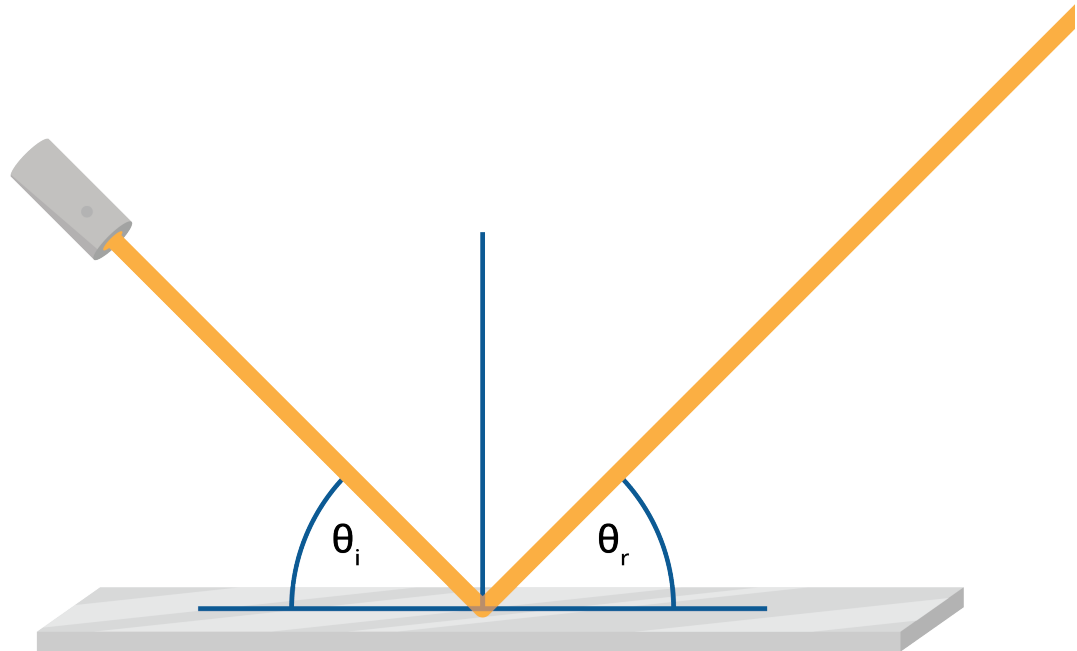


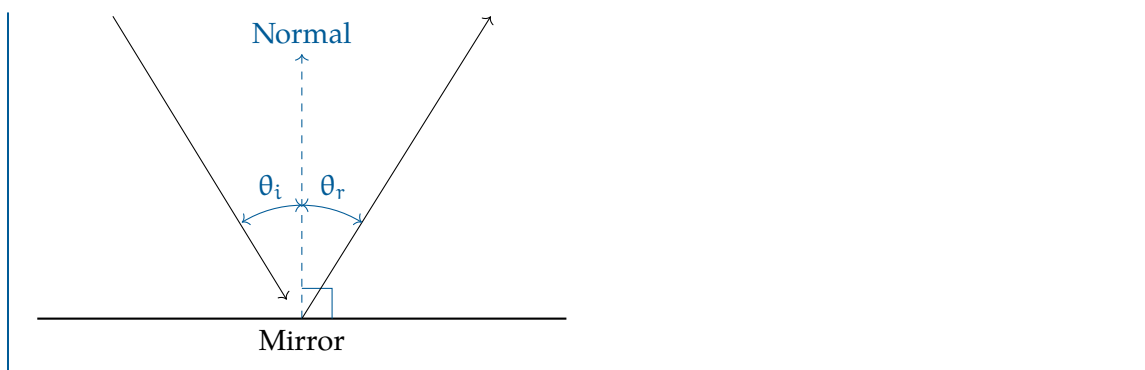
Figure 2.1: The angle of *incidence* is equal to the angle of *reflection*.

#### Law of Reflection

The angle of incidence, denoted as  $\theta_i$ , is equal to the angle of reflection, denoted as  $\theta_r$ . This law can be mathematically expressed as:

$$\theta_i = \theta_r$$

where  $\theta_i$  is the angle between the incident light ray and the normal to the surface, and  $\theta_r$  is the angle between the reflected light ray and the normal.



### Exercise 2 Law of Reflection

Working Space

You are standing 4 meters from a mirror hung on a wall. The bottom of the mirror is the same height as your chin, so you can't see your whole body. You stick a piece of masking tape to your body.

You walk forward until you are only 3 meters from the mirror, then put a piece of masking tape on your body at the new cut-off point. Is the new masking tape higher or lower on your body?

Answer on Page 75

**Exercise 3      Photons and Color***Working Space*

There are red photons.

Are there black photons?

Are there white photons?

Are there yellow photons?

*Answer on Page 75***2.3    Ellipses and Curved Mirrors**

Flat mirrors are common and useful, but things get more interesting once you bend the mirrors. In this section, we are going to talk about a few different kinds of curved mirrors.

**2.3.1    The Reflective Properties of Circles and Spheres**

For example, if you were inside a circular room (a cylinder, actually), you could imagine standing in the center and pointing a flash light in any horizontal direction. The beam of light would bounce right back to you. See Figure 2.2

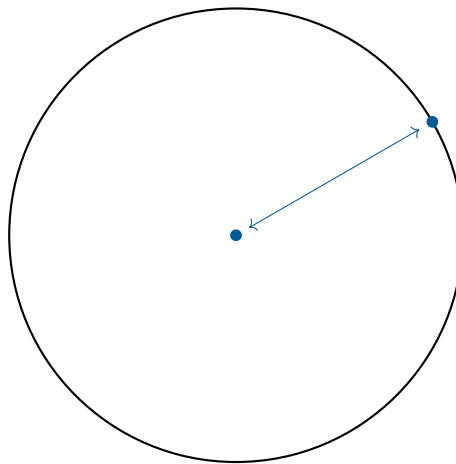


Figure 2.2: A circle would reflect back onto itself.

How do you know this? Because the tangent line is always perpendicular to the radius to the point of tangency:

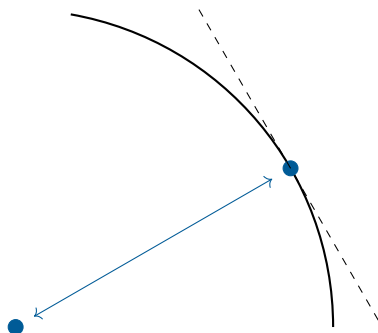


Figure 2.3: The reflection on a circle uses a radius which hits a tangent line as a perpendicular bisector.

You could create a spherical room with mirror walls. You would create a platform in the center where you could stand, and if you pointed your flashlight in any direction, its beam of light would shine back at you.

### 2.3.2 Ellipses and Ellipsoids

Intuitively, you know what an ellipse is: an oval. However, the ellipse is actually an oval with some special properties. This is a good time to talk about those properties.

Mathematicians talk about a *standard* ellipse. A standard ellipse is centered on the origin  $(0, 0)$  and its long axis is parallel with the  $x$ -axis or the  $y$ -axis.

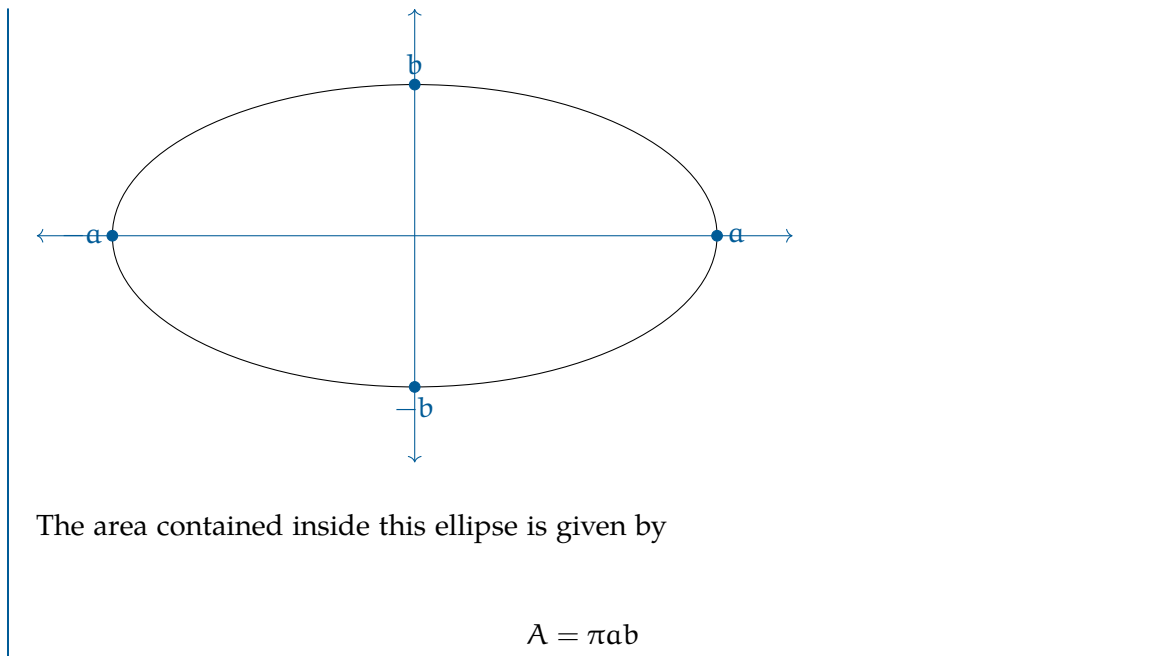
#### Equation for a Standard Ellipse

To be precise, a standard ellipse is the set of points  $(x, y)$  that are solutions to the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Note that  $(a, 0)$ ,  $(-a, 0)$ ,  $(0, b)$ ,  $(0, -b)$  are all part of the set. The complete set looks like this:





We can now talk about two special points: the *foci*. Each focal point is on the long axis of the ellipse. Let's assume for a moment that  $a > b$ . (Everything works the same if  $b > a$ , but it gets confusing if we try to deal with both cases simultaneously.)

If  $p$  is a point on the ellipse, the distance from  $p$  to focal point 1 plus the distance from  $p$  to focal point 2 is always  $2a$ .

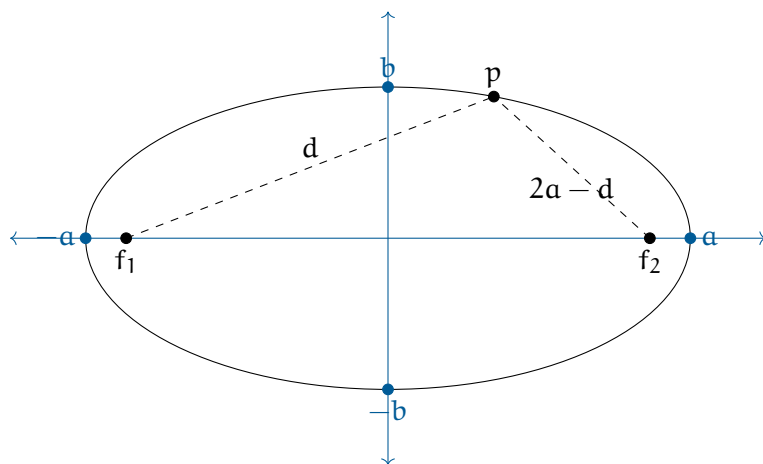


Figure 2.4: The both foci and point  $p$  forms a triangle with legs  $2a - d$  or  $d$ .

How do we find the foci? We know they are on the long axis and that they are symmetrical across the short axis. All we need to know is how far are they from the short axis.

**Distance from Center to the Foci**

If you have an ellipse with a long axis that extends  $a$  from the center, and a short axis that extends  $b$  from the center, the foci lie on the long axis and are  $c$  from the center. Where

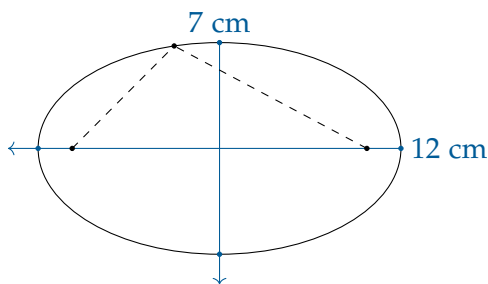
$$c = \sqrt{a^2 - b^2}$$

**Exercise 4      Foci of an ellipse**

*Working Space*

You need to draw an ellipse that is 12 cm long and 7 cm wide. You have a string, two pushpins, a ruler, and a pencil. Using the ruler, you draw two perpendicular axes.

You will stick one pin at each focal point. Each end of the string will be tied to a push pin. Using the pencil to keep the string taut, you will draw an ellipse.



How far from the short axis are the pushpins placed?

How long is the string between them?

*Answer on Page 76*

### The Reflective Property of Ellipses

Here is something else that is wonderful about an ellipse: Pick any point  $p$  on the ellipse. Draw a line from  $p$  to each focal point. Draw the line tangent to the ellipse at  $p$ . You will see that the angle between the tangent and the line to focal point 1 is equal to the angle between the tangent and the line to focal point 2. See Figure 2.5

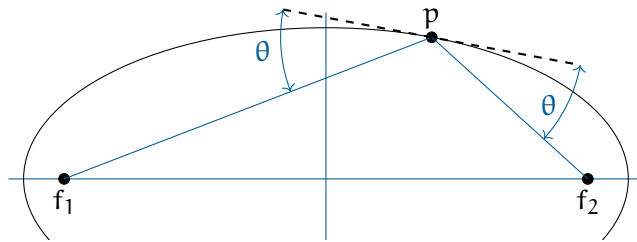


Figure 2.5: Reflective Property of Ellipses.

This is known as “The Reflective Property of Ellipses”. Imagine you and your friend Fred are at an ellipse-shaped skating rink, and the edge of the rink is mirrored. You sit at one focal point and your friend sits at the other. If you point a flashlight at the mirror (in any direction!), the beam will bounce off the wall and head directly for Fred.

If Fred ducks out of the way, the beam will bounce again and head back to you.

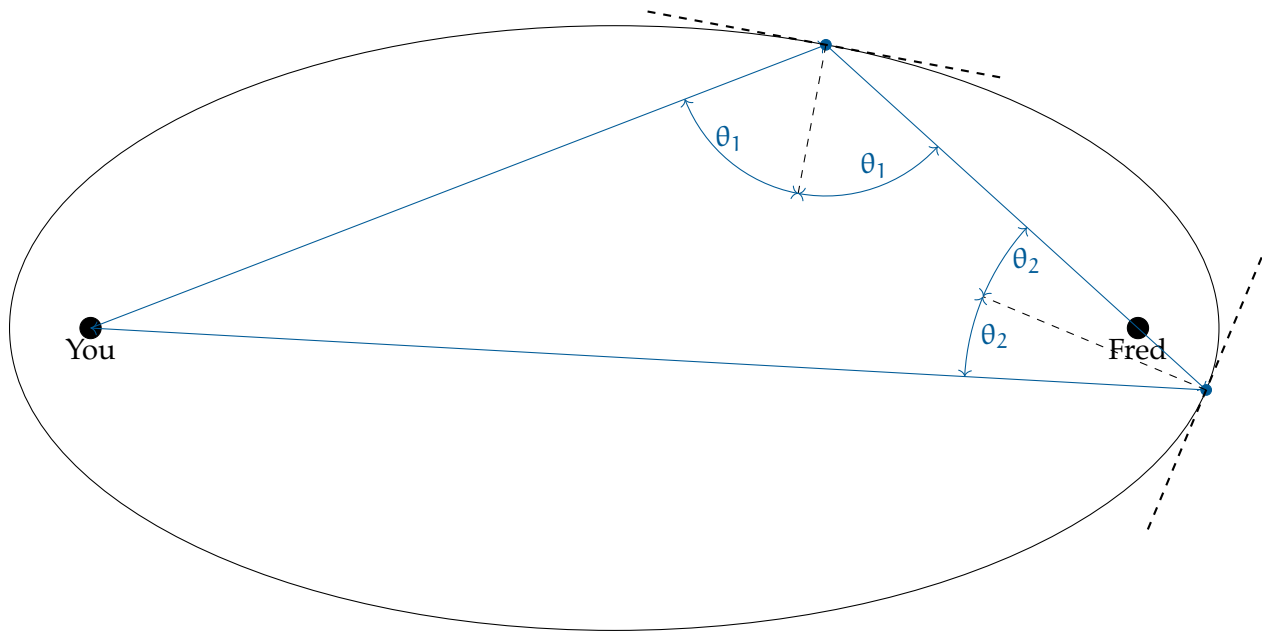


Figure 2.6: Your laser intersects Fred.

This will work for sound as well. If you whisper while on the focal point, Fred (at the other focal point) will hear you surprisingly well, because all the soundwaves that hit the wall will bounce (just like the light) straight at Fred.

### 2.3.3 Elliptical Orbits

One more fun fact about ellipses: We often imagine the planets traveling in circular orbits with the sun at the center — they actually travel in elliptical orbits, with the sun as one of the focal points.

The Earth is closest to the sun around January 3rd: 147 million km.

The Earth is farthest from the sun around July 3rd: 152 million km.

(Note that these dates are not the same as the solstices: The southern hemisphere is tilted the most toward the sun around December 21 and tilted most away around June 21.)

### 2.3.4 Ellipsoids

Just as we can pull the ideas of a circle into three dimensions to make a sphere, we can extend the ideas of the ellipse into three dimensions to talk about ellipsoids. Ellipsoids are like blimps.

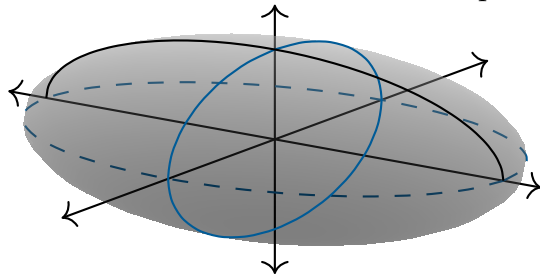
The standard ellipsoids are centered at the origin and aligned with the three axes.

#### Equation for a Standard Ellipsoid

To be precise, a standard ellipse is the set of points  $(x, y, z)$  that satisfy the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Note that  $(a, 0, 0)$ ,  $(0, b, 0)$ ,  $(0, 0, c)$  are all part of the set. The complete set looks like



this:

The volume bounded by this ellipsoid is

$$V = \frac{4}{3}\pi abc$$

Of course,  $a$ ,  $b$ , and  $c$  can be any positive number, but in the real world, we find ourselves working regularly with ellipsoids where two of the numbers are the same.

#### Oblate Spheroid

If two axes have the same length and one is shorter, you get something that looks like a sphere compressed in one direction — like a pumpkin. These are called *oblate spheroids*.

The Earth is actually an oblate spheroid; the axis that goes through the north and south pole is shorter than the axes that pass through the equator. How much shorter? Just a little, relatively speaking. The equator is 6,378 km from the center of the Earth; the north pole is 21 km closer.

#### Prolate Spheroid

If two axes have the same length and one is longer, you get something that looks like a sphere stretched in one direction — like a rugby ball. It is called a *prolate spheroid*.

Like an ellipse, prolate spheroids have two focal points.

**Focal Points of a Prolate Spheroid**

If the long axis has a radial length of  $a$  and the two shorter axes have radial length  $b$ , then the focal points are on the long axis. The distance from the center to the focal point is given by

$$c = \sqrt{a^2 - b^2}$$

For any point  $p$  on the prolate spheroid, the sum of the distances from  $p$  to the focus points will always be  $2a$ .

It has the reflective property: A photon shot in any direction from one focal point will bounce off the wall and head directly at the other.

**Exercise 5 Volume of Ruby Ball**

*Working Space*

Some jokesters once thought it would be fun to make something that looked like a rugby ball, but made out of lead.

A rugby ball is about 30 cm long and has a circumference of 60 cm at its midpoint. A cubic centimeter of lead has a mass of 11.34 grams.

How much would a solid (not hollow) lead ruby ball weigh?

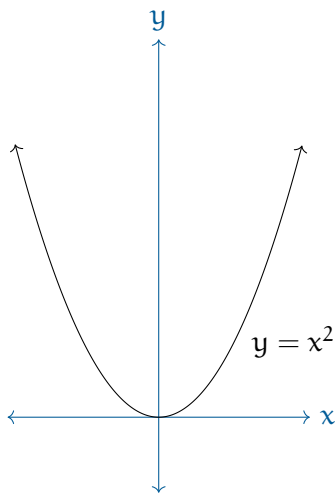
*Answer on Page 76*

**2.3.5 Parabolas and Parabolic Reflectors**

You are familiar with quadratic functions:

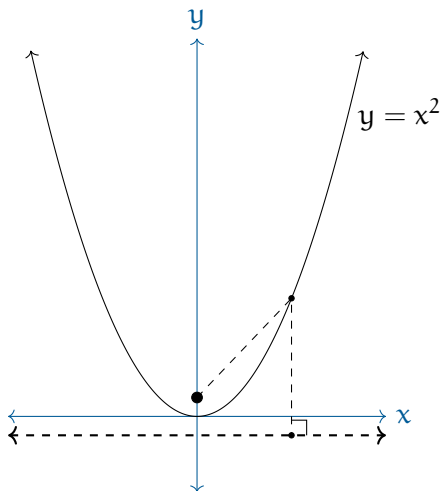
$$y = ax^2 + bx + c$$

If  $a$  is not zero, the graph of a quadratic is a curved line called a *parabola*. The first parabola that most mathematicians think of is the graph of  $y = x^2$ :



Every parabola has a *focus* and a *directrix*. The focus is a point on the parabola's axis of symmetry. The directrix is a line perpendicular to the axis of symmetry. Every point on the parabola is equal distance from the focus and the directrix.

For the graph of  $y = x^2$ , the focus is the point  $(0, \frac{1}{4})$ . The directrix is the line  $y = -\frac{1}{4}$ :



For example, the point  $(1, 1)$  is on this parabola. It is  $5/4$  from the directorix. How far is it from the focus? 1 horizontally and  $3/4$  vertically.

$$\sqrt{1^2 + \left(\frac{3}{4}\right)^2} = \frac{5}{4}$$

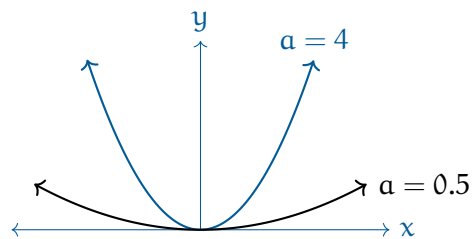
Thus, we have confirmed that  $(1, 1)$  is equal distances from the focus and the directrix.

When we think about a parabola and its properties, we usually rotate and translate it to be symmetric around the  $y$ -axis, flip it so that it is low in the middle and rising on both sides, and push it up or down until the low point is on the  $x$ -axis.

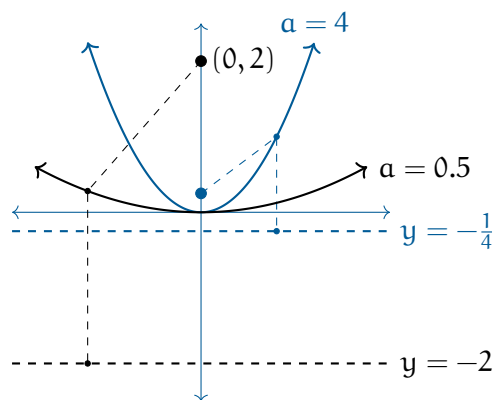
Then, they can all be written:

$$y = \frac{a}{4}x^2$$

where  $a > 0$ . If  $a$  is small, the parabola opens wider.



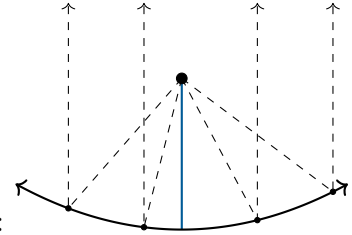
The focus is at  $(0, \frac{1}{a})$  and the directorix is the line  $y = -\frac{1}{a}$ .





**Reflective Property of a Parabola**

Assume you have a parabola-shaped mirror. A beam of light shot from the focus will



bounce off the mirror in the direction of the axis of symmetry:

This is why your flashlight has a parabolic mirror. The lightbulb is at the focus, so any photons that hit the mirror are redirected straight forward.

(Note that in the real world, we use parabolic dishes: a parabola rotated around its axis of symmetry.)

The reflection works exactly the same in reverse. There are solar cookers that are big parabolic mirrors. They let you put a pot on the focus point. You move the dish until its axis of symmetry is pointed at the sun.

You will also see a lot of antennas have parabolic dishes. Note that photons that come in parallel to the axis of symmetry are redirected to a single point: where the receiver is.



Figure 2.7: A real life parabolic dish as a satellite.

Sometimes in a science museum, you will see two parabolic dishes far apart and pointed at each other. One person speaks with their mouth at the focus of one. The other person listens with their ear at the focus of the other. Even though you are very far apart, it sounds like they are really, really close.



Figure 2.8: Two parabolic dishes send waves between each other.

This is because the speaker's parabolic wall focuses the sound energy in a nice beam the size of the wall pointed straight at the listener's parabolic wall. The listener's wall focuses

the energy of that beam at the listener's ear.



# Refraction

The refraction of light is a phenomenon where light changes its direction when it passes from one medium to another. The change in direction is due to a change in the speed of light as it moves from one medium to another.

This phenomenon is explained by Snell's law, which states:

$$n_1 \cdot \sin(\theta_1) = n_2 \cdot \sin(\theta_2) \quad (3.1)$$

where:

- $n_1$  and  $n_2$  are the indices of refraction for the first and second media, respectively. The index of refraction is the ratio of,  $c$  the speed of light in a vacuum to,  $v$  the speed of light in the medium,  $n = \frac{c}{v}$ . It is a dimensionless quantity.  $n \geq 1$  for all materials, such that  $n_{\text{vacuum}} = 1$  and  $n_{\text{air}} \approx 1$ . See a list of all mediums here: [https://en.wikipedia.org/wiki/List\\_of\\_refractive\\_indices](https://en.wikipedia.org/wiki/List_of_refractive_indices)
- $\theta_1$  and  $\theta_2$  are the angles of incidence and refraction, respectively. These angles are measured from the normal (perpendicular line) to the surface at the point where light hits the boundary.

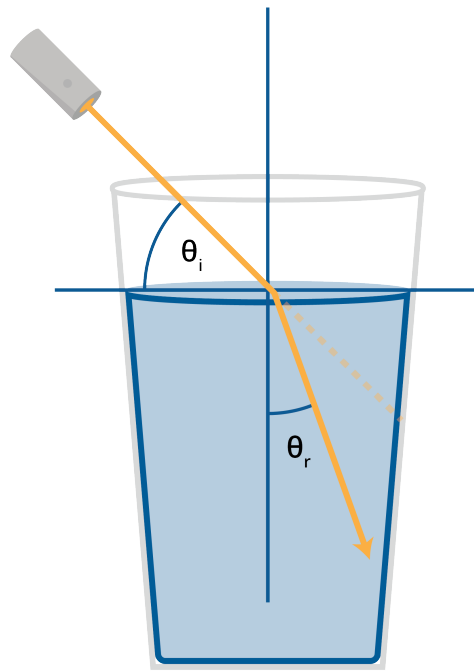


Figure 3.1: Refraction occurs when light changes the medium it is in.

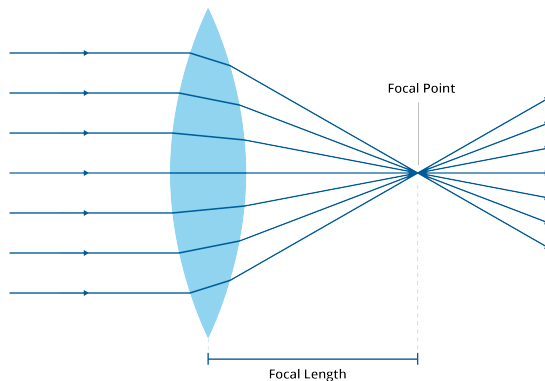
The angle of incidence ( $\theta_1$ ) is the angle between the incident ray and the normal to the interface at the point of incidence. Similarly, the angle of refraction ( $\theta_2$ ) is the angle between the refracted ray and the normal.

When light travels from a medium with a lower refractive index to a medium with a higher refractive index, it bends towards the normal. Conversely, when light travels from a medium with a higher refractive index to one with a lower refractive index, it bends away from the normal.

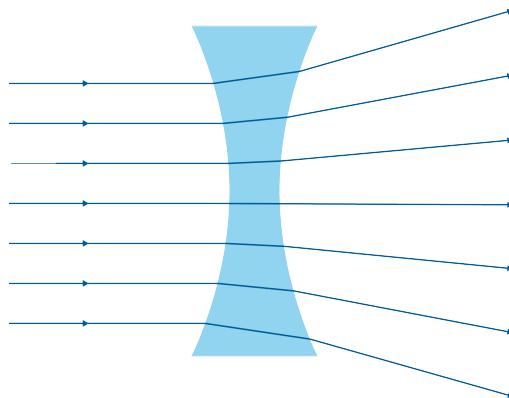
# Lenses

Lenses are optical devices with perfect or approximate axial symmetry that transmit and refract light, converging or diverging the beam. There are two main types of lenses, distinguished by their shape and the way they refract light:

- **Converging (or Convex) Lenses:** These are thicker at the center than at the edges. When parallel light rays enter a convex lens, they converge to a point called the focal point. Examples of converging lenses include magnifying glasses and camera lenses.



- **Diverging (or Concave) Lenses:** These are thinner at the center than at the edges. When parallel light rays enter a concave lens, they diverge or spread out. These lenses are often used in glasses to correct nearsightedness.



## 4.1 Focal Length

The focal length of a lens is the distance between the center of the lens and the focal point. It is determined by the lens shape and the refractive index of the lens material. For a converging lens, the focal length is positive; for a diverging lens, the focal length is negative.

## 4.2 Refractive Index

The refractive index of a material is a measure of how much the speed of light is reduced inside the material. The refractive index  $n$  of a material is given by the ratio of the speed of light in a vacuum  $c$  to the speed of light  $v$  in the material:

$$n = \frac{c}{v}$$

The refractive index affects how much a light ray changes direction, or refracts, when entering the material at an angle. A higher refractive index indicates that light travels slower in that medium, and the light ray will bend more towards the normal.

Lenses work by refracting light at their two surfaces. By choosing the right lens shape and material, lenses can be designed to bring light to a focus, spread it out, or perform more complex transformations.



# Oscillations

We just finished up a chapter about circular motion, where objects move around in a circle at either constant (uniform/harmonic) or changing (non-uniform) speeds.

In this chapter, we will study oscillatory motion, which is a type of motion where an object moves back and forth around an equilibrium point. Oscillations are common in many physical systems, such as springs, pendulums, and even electrical circuits. Oscillations are important not only because they appear everywhere in physical systems, but also because they provide a connection between simple motion and more complex behaviors. Small oscillations often behave in a very predictable way and can be described using *simple harmonic motion* (SHM), which is governed by restoring forces proportional to displacement.

We will also give an introduction to *differential equations* used to model oscillatory systems, often found in engineering and physics.

## 5.1 Sine and Cosine Graphs [Review]

Let's recall the two pillars of trigonometry and, consequently, simple harmonic motion: Sine and Cosine. Recall that these two functions are periodic, meaning they repeat their values in regular intervals or periods. The sine and cosine functions have a period of  $2\pi$  radians or  $360^\circ$ , repeating their values every  $2\pi$  radians.

As discussed in the circular motion chapter, the sine function represents the y-coordinate of a point on the unit circle as it moves around the circle, while the cosine function represents the x-coordinate. This is the fundamental behavior of the unit circle, where a point moves around the circle at a constant angular speed  $\omega$ . The position of the point at any time  $t$  can be described using the following parametric equations:

$$x(t) = \cos(\omega t), \quad y(t) = \sin(\omega t)$$

The general forms of the sine and cosine functions can be expressed as:

$$x(t) = A \cos(\omega t + \phi), \quad y(t) = A \sin(\omega t + \phi)$$

where:

- $A$  is the amplitude, representing the maximum displacement from the equilibrium position.
- $\omega$  is the angular speed, which determines how quickly the oscillations occur.
- $\phi$  is the phase shift, which determines the initial position of the oscillation at  $t = 0$ .

Recall that the derivatives of these functions are also periodic, a key property that is useful for analyzing oscillatory motion:

$$\frac{d}{dt} \cos(\omega t) = -\omega \sin(\omega t)$$

$$\frac{d}{dt} \sin(\omega t) = \omega \cos(\omega t)$$

$$\frac{d^2}{dt^2} \cos(\omega t) = -\omega^2 \cos(\omega t)$$

In the motion we will talk about,

- $x(t)$  will be our position function
- $v(t)$  describes our velocity function, which can be the first derivative of our position
- $a(t)$  is our acceleration function, derived from  $v(t)$

We notice that the second derivative produces a negative multiple of the original function. This can be simplified in terms of acceleration and position functions:  $a(t) = -\omega^2 x(t)$ .

## 5.2 Simple Harmonic Motion

Simple Harmonic Motion will reference many definitions we have seen before and will continue to see in this chapter. Here are some key terms to know:

**Cycle** A single complete execution of a periodically repeated motion. Another way to think about it is a particle, block, or object making a *round-trip*.

**Period** The time it takes to complete one cycle of motion. An identity that comes up often is  $T = \frac{2\pi}{\omega}$ .

**Frequency** The number of cycles completed per unit time. It is the reciprocal of the period, represented as **Hz**.<sup>1</sup>

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<sup>1</sup>One **Hz** is equal to one cycle per second. For example, a frequency of 5 Hz means that 5 cycles are completed in one second. On a monitor, a refresh rate of 60 Hz means the screen updates 60 times per second. Higher refresh rates can lead to smoother motion perception.

**Equilibrium** The position at which the “default” position of an oscillating object is located. In terms of mechanics, this is the position where all the forces acting on the object are balanced or zero.

**Restoring Force** The force that acts to bring an oscillating object back to its equilibrium position. It is typically proportional to the displacement from equilibrium and acts in the opposite direction.

**Angular Speed** The rate at which an object moves around a circle, measured in radians per second.

So far, we have discussed oscillatory motion in general terms. Now, let’s focus on a specific type of oscillatory motion known as Simple Harmonic Motion (SHM). SHM is characterized by the following properties:

- The motion occurs around an equilibrium position.
- The *restoring force* is directly proportional to the displacement from the equilibrium position and acts in the opposite direction.
- The motion is sinusoidal in nature, meaning it can be described using sine or cosine functions.
- The period and frequency of the motion are constant, regardless of the amplitude.
- The acceleration always points toward equilibrium and increases in size as the displacement grows.

So the motion continuously repeats itself in a regular pattern. It speeds up near equilibrium, and slows down to a stop at maximum displacement. Examples of these systems are

- A mass attached to a spring, either vertically or horizontally
- A simple pendulum
- Vibrations of a tuning fork or guitar string

We saw in our pendulums chapter how the system has both centripetal acceleration and acceleration due to gravity, causing oscillations back and forth with changing speed. Like all mechanical systems, an SHM system has both kinetic and potential energy. For general simple harmonic motion systems, we can describe its energy as follows:

- The total mechanical energy  $E$  of the system is the sum of its kinetic energy  $K$  and potential energy  $U$ :

$$E = K + U$$

- The kinetic energy  $K$ , system dependent, is highest when the object passes through

the equilibrium position.

- The potential energy  $U$  depends on the specific system, such as spring potential energy or gravitational potential energy, but is highest at the maximum displacement from equilibrium.

## 5.3 Springs

Let's talk about springs! Springs are mechanical devices, usually made out of some rigid metal material, that store and release energy through deformation. When a spring is stretched or compressed from its equilibrium position, it exerts a restoring force that tries to bring it back to its original shape. The most common type of spring is a coil spring, which consists of a linear wire wound into a helical shape.

### 5.3.1 Hooke's Law

In 1676, physicist Robert Hooke formulated a principle that describes the behavior of springs, now known as Hooke's Law. Hooke's Law states that the force exerted by an ideal spring is directly proportional to the displacement from its equilibrium position.

Mathematically, Hooke's Law can be expressed as:

$$F_s = -kx$$

where:

- $F_s$  is the restoring force exerted by the spring (in newtons, N)
- $k$  is the spring constant (in newtons per meter, N/m), which measures the stiffness of the spring. It is spring-specific
- $x$  is the displacement from the equilibrium position (in meters, m)
- The negative sign represents the fact that the force attempts to bring the spring back to its equilibrium position.

Since we know that force is related to mass and acceleration through Newton's Second Law, we can combine Hooke's Law with Newton's Second Law to analyze the motion of a mass attached to a spring. See Figure 5.1 for a diagram of a mass-spring system.

$$F_s = ma = -kx \tag{5.1}$$

$$ma = -kx \tag{5.2}$$

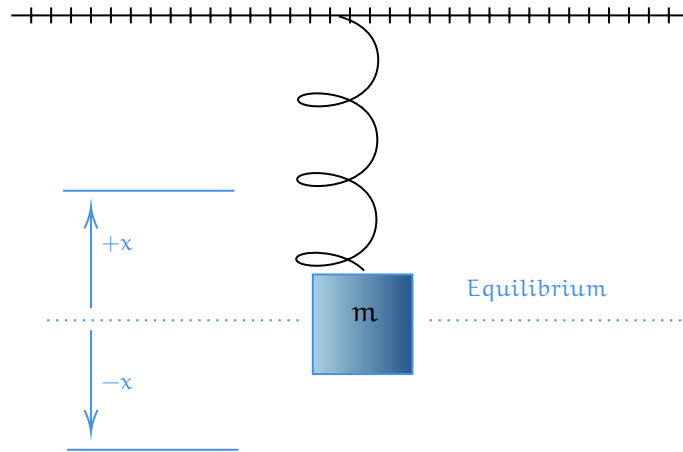


Figure 5.1: A diagram showing a block of mass  $m$  displaced  $+x$  and  $-x$  from equilibrium.

But since above we found a relation between acceleration and position, we can write:

$$m \frac{d^2x}{dt^2} = -kx \quad (5.3)$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad (5.4)$$

We can see that this equation describes simple harmonic motion, where the acceleration is proportional to the negative of the displacement. In this form, we can identify the angular speed  $\omega$  of the oscillation as:

$$\omega = \sqrt{\frac{k}{m}}$$

as the term in front of the  $x$  in equation (5.4) is  $\omega^2$ . Thus, we can further define the period as

$$T = 2\pi\sqrt{\frac{m}{k}}$$

and the frequency as

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

In a later section of this chapter, we will solve this differential equation to find the position function  $x(t)$  of general mass-spring systems.

### Exercise 6 Mass-Spring Frequency Change

A block is attached to a spring and set into oscillatory motion, and its frequency is measured. If this block were removed and replaced with a block of  $1/2$  the mass, how would the frequency of oscillations compare to the original frequency?

*Working Space*

1. The frequency would be halved.
2. The frequency would remain the same.
3. The frequency would increase by a factor of  $\sqrt{2}$ .
4. The frequency would decrease by a factor of  $\sqrt{2}$ .

*Answer on Page 76*

### 5.3.2 Spring Potential Energy

Let's recall that Work from a physics standpoint is defined as the transfer of energy through a certain distance via a force. When we alter the length of a spring through either compression or extension, we are performing work on the spring. This work is stored as potential energy within the spring, which can be released when the spring returns to its equilibrium position.

Let's find an equation for this change in potential energy, which gets stored up in the spring. Recall the two mathematical definitions for work:

- Work as the product of force and distance:  $W = F \cdot d$
- Work as the integral of force over a distance:  $W = \oint F dx$

We need to use the integral definition, since, by our definition of Hooke's Law, the force exerted by a spring varies with displacement, so every tiny bit of distance, or  $dx$  is different. For any spring starting at equilibrium (0), and moving to some final position  $x_f$ , we

can write:

$$\begin{aligned}
 W &= \int_{x_i}^{x_f} F_x \, dx \\
 W_s &= \int_{x_i}^{x_f} F_s \, dx \\
 &= \int_0^{x_f} -kx \, dx \\
 &= \left. \frac{-kx^2}{2} \right|_0^{x_f} \\
 &= \frac{-k(x_f)^2}{2}
 \end{aligned}$$

This work done on the spring is stored as potential energy, so we can say that the change in potential energy of the spring is equal to the negative of the work done by the spring force:

$$\Delta U_s = -W_s = \frac{1}{2}kx_f^2 \quad (5.5)$$

We can see this graphically in Figure 5.2, where the area under the force vs. displacement graph represents the work done on the spring, which is equal to the change in potential energy.

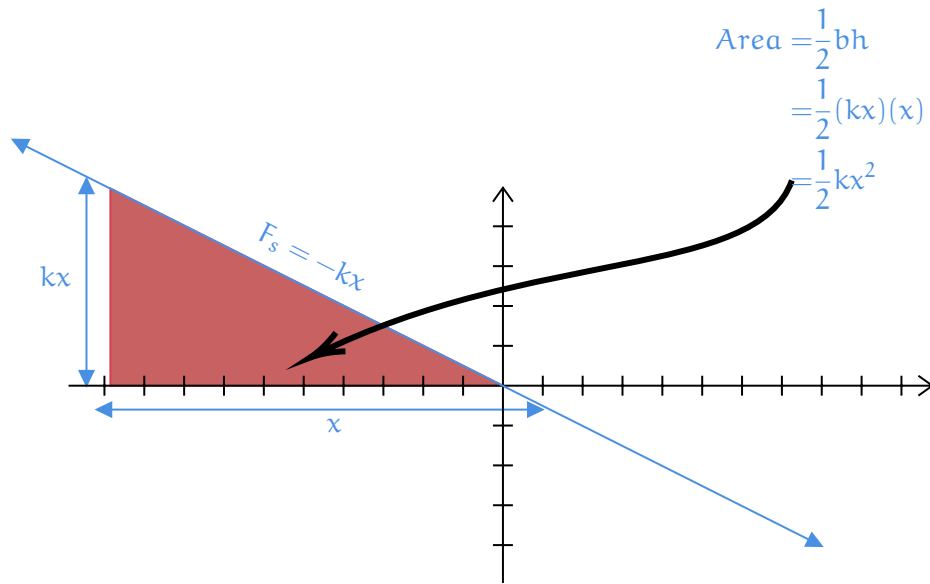


Figure 5.2: A graph of force versus displacement for a spring, equal to the work done and change in potential energy.

Notice that Equation (5.5) is similar in structure to the kinetic energy equation,  $K = \frac{1}{2}mv^2$ . This is not a coincidence, as both equations describe forms of energy in mechanical systems, and are derived from linear systems. When the effort (force) needed increases in

direct proportion to what you're trying to change, the energy required ends up depending on the square of that change. For a spring

- the farther you stretch or compress it, the more force it exerts to return to equilibrium (Hooke's Law)
- the more energy is stored in the spring (spring potential energy)

For kinetic energy

- the faster you try to move an object, the more force is needed to accelerate it (Newton's Second Law)
- the more energy the object has due to its motion (kinetic energy)

### Exercise 7 Finding the equation for a mass-spring system

This question is from an AP-style physics review. A block of mass 4 kg on a frictionless, horizontal table is attached to a spring with spring constant  $k = 400 \text{ N/m}$  and undergoes simple harmonic oscillations about its equilibrium position and its amplitude  $A = 6 \text{ cm}$ . If the block is at  $x = 6 \text{ cm}$  at time  $t = 0$  and has a velocity of  $0 \text{ cm/s}$  at that time, which of the following equations (with  $x$  in cm and  $t$  in seconds) describes the block's position as a function of time?

Working Space

- (A)  $x(t) = 6 \cos(10t)$
- (B)  $x(t) = 6 \sin(10t)$
- (C)  $x(t) = 6 \sin(10t + \frac{1}{2}\pi)$
- (D)  $x(t) = 6 \sin(10\pi t + \frac{1}{2}\pi)$
- (E)  $x(t) = 6 \sin(10\pi t - \frac{1}{2}\pi)$

Answer on Page 77



### Exercise 8      Energy Transfer in a Mass-Spring System

A pinball machine uses a spring to launch a ball of mass 0.2 kg. The spring has a spring constant of 500 N/m and is compressed by 0.24 m before release. Assuming energy is conserved and no frictional losses occur, what is the speed of the ball as it leaves the spring?

Working Space

Answer on Page 78

## 5.4 Mass-Spring Systems and Linear Differential Equations

We have already seen that the motion of a mass attached to a spring can be modeled using differential equations. Here, we will examine this connection more closely.

### 5.4.1 Undamped Simple Harmonic Motion

Starting from Newton's Second Law and Hooke's Law, we derived the differential equation for simple harmonic motion:

$$m \frac{d^2x}{dt^2} + kx = 0. \quad (5.6)$$

The term  $m \frac{d^2x}{dt^2}$  comes directly from *Newton's Second Law*, representing the net force required to accelerate the mass. The term  $kx$  arises from *Hooke's Law*, which states that the spring exerts a restoring force proportional to displacement and directed toward equilibrium. Equation (5.6) is an example of a **second-order linear differential equation with constant coefficients without damping**. It describes an ideal mass-spring system with no external influences: one that oscillates forever, undamped and unforced. The  $ma(t)$  term is *Newton's Second Law* part of the equation, while the  $kx(t)$  term comes from *Hooke's Law*. This is a second-order linear differential equation with constant coefficients. What if the spring system is being driven by a third force, like friction or damping? We can add an additional force term  $F_f(t)$  to the equation:  $F_f(t) = -cv = -cx'$  for forcing function.

### 5.4.2 Introducing Damping

Real systems are often influenced by additional forces. One common example is a **damping force**, such as friction or air resistance, which opposes the motion of the mass. A typical damping force is proportional to the velocity and can be written as

$$F_d(t) = -c \frac{dx}{dt},$$

where  $c$  is the damping coefficient. Including this force in Newton's Second Law modifies our differential equation. Instead of the undamped equation (5.6), we obtain:

$$m\ddot{x} = -c\dot{x} - kx(t) \quad (5.7)$$

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0 \quad (5.8)$$

This is still a second-order linear differential equation with constant coefficients, but its solutions behave very differently. Depending on the value of  $c$ , the system may oscillate with decreasing amplitude, fail to oscillate at all, or return to equilibrium as quickly as possible. Finding the roots of the characteristic equation associated with Equation (5.8) allows us to classify the system's behavior into three categories. Recall that the quadratic formula gives us the roots using our coefficients:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

| Characteristic Roots                      | Homogeneous Solution                                         | Damping Type           |
|-------------------------------------------|--------------------------------------------------------------|------------------------|
| $r_1, r_2 \in \mathbb{R}$ , both negative | $y_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$                        | Overdamped             |
| $r$ real, repeated                        | $y_h = (C_1 + C_2 t) e^{rt}$                                 | Critically damped      |
| $r = \alpha \pm i\beta$ , $\alpha < 0$    | $y_h = e^{\alpha t} (C_1 \cos(\beta t) + C_2 \sin(\beta t))$ | Underdamped            |
| $r = \pm i\beta$                          | $y_h = C_1 \cos(\beta t) + C_2 \sin(\beta t)$                | Simple Harmonic Motion |

Table 5.1: Classification of solutions to the damped harmonic oscillator based on characteristic roots.

Without going into too much linear algebra, the function of a mass spring is given by  $y(t) = y_h(t) + y_p(t)$ , where  $y_h$  is the homogeneous solution (the part we solved above by setting the differential equation to 0), and  $y_p$  is the particular solution, which depends on any external forcing functions. In this section, we focused on the homogeneous solution. If the equation (5.8) had a forcing function  $F(t)$  on the right side, we would need to find a particular solution  $y_p$  to account for that.

**Exercise 9      Linear Systems 1***Working Space*

A 2 kg mass is attached to a spring with  $k = 200 \text{ N/m}$  with no damping. Which differential equation describes the motion?

1.  $x'' + 100x = 0$
2.  $x'' + 10x = 0$
3.  $x'' + 200x = 0$
4.  $2x'' + 100x = 0$

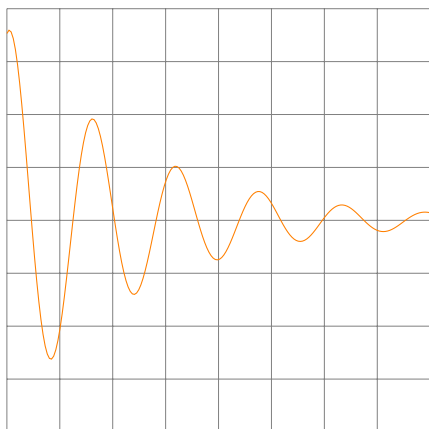
*Answer on Page 78*

**Exercise 10**      **Linear Systems 2**

Work

Answer the following questions about the graph below, which shows the position of a mass on a spring as a function of time. The equation of motion for this system is given by:

$$x(t) = 0.15e^{-0.40t} \cos(4t + 0.30)$$



- Is this system undergoing simple harmonic motion, or is it damped? If it is damped, is it underdamped, overdamped, or critically damped?
- What would need to be changed in the equation of motion to make this system undergo simple harmonic motion instead?

Answer on Page 78

**Exercise 11      Linear Systems 3***Working Space*

Given that a unit (1 kg) mass obeys:

$$x'' + 16x = 0, \quad x(0) = 0.20, \quad v(0) = x'(0) = 0$$

1. Solve for  $x(t)$
2. Find the acceleration,  $a(t)$ , and the velocity,  $v(t)$ , as functions of time.

*Answer on Page 78***5.5 Pendulums**

You may be surprised to know that pendulums act the exact same way as roller coasters from the circular chapter, in terms of forces. But, similar to springs, they oscillate back and forth unless acted upon by a damping force. A pendulum has an equilibrium position  $\theta = 0$  such that it is parallel to its gravitational component. A pendulum also has a maximum angle  $\theta_{\max}$  from its equilibrium position. It will swing through a set arc in a repetitive motion, which is a type of motion we call simple harmonic motion, similar to springs (which we will also cover in the oscillations chapter). The tension force is always directed towards the pivot. This is diagramed below in Figure 5.3

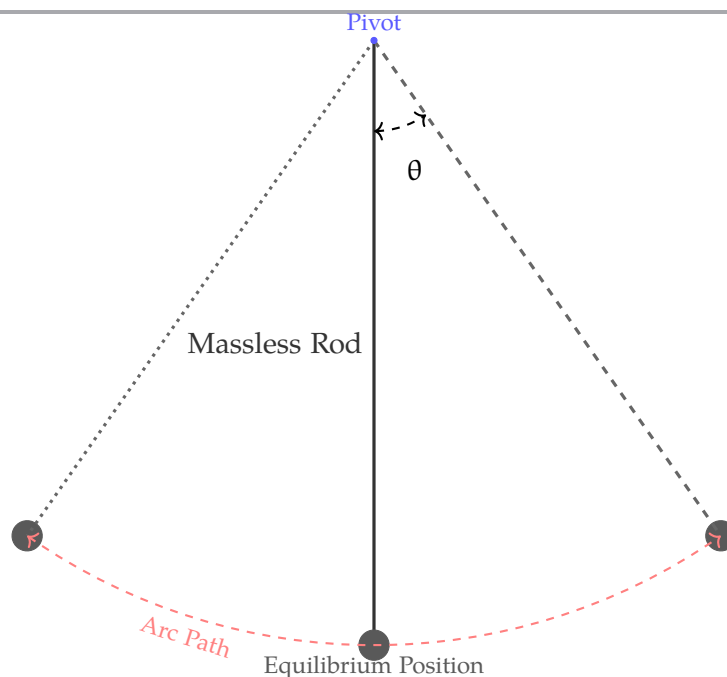


Figure 5.3: The anatomy of a pendulum.

When displaced by an angle  $\theta$  from the vertical equilibrium position ( $\theta = 0$ ), gravity provides a *restoring force* that acts along the arc of motion. This restoring force is given by  $F = -mg \sin \theta$ , where the negative sign indicates that the force always points back toward the equilibrium position. The tension in the string keeps the bob moving along its curved path, while gravity continually tries to return it to rest. The net force shifts such that the acceleration arrow is *not* always pointing to the center. The net force is a combination of the tension (centripetal) force and the gravitational (tangential), so rather it points at different points along the arc of the circle, as it is always changing due to the height of the pendulum. This is demonstrated in Figure 5.4. Note the following differences in energy and forces at the equilibrium position versus the maximum angle positions, summarized in Table 5.2. Note here that the magnitude of the restoring

| At $\theta = 0$                     | At $\theta = \theta_{\max}$   |
|-------------------------------------|-------------------------------|
| $F_{\text{restoring}} = 0$          | $F_{\text{restoring}} = \max$ |
| $a_t$ (tangential acceleration) = 0 | $a_t = \max$                  |
| PE = 0                              | PE = mgh                      |
| KE = max                            | KE = 0                        |
| V = max                             | v = 0                         |

Table 5.2: Comparison of pendulum quantities at equilibrium and maximum displacement

force— $mg \sin(\theta)$ , which is not *directly* proportional to the angular displacement  $\theta$ . However, for small angles (typically less than about 1 radian or  $180^\circ$ ), we can use the small-angle approximation  $\sin(\theta) \approx \theta$  (in radians), and state the restoring force as  $F \approx -mg\theta$ . This allows us to treat the pendulum's motion as simple harmonic motion for small oscillations, where the restoring force is approximately proportional to the angular displace-

ment.

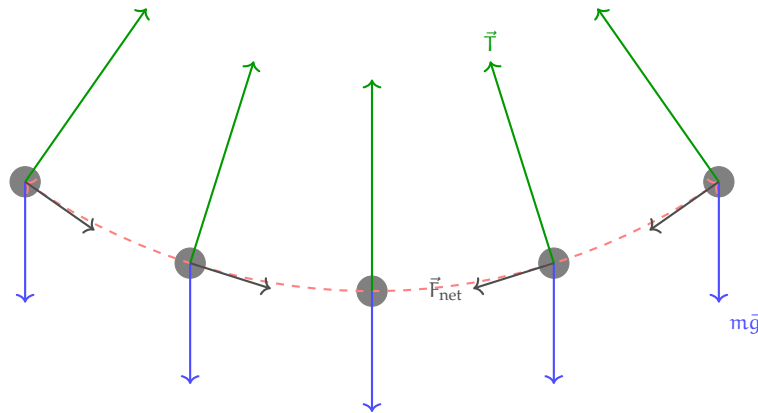


Figure 5.4: The net force at different locations of the arc.

### Exercise 12 Pendulum Period Dependence

A simple pendulum consists of a mass  $m$  attached to a massless rod  $L$ , swinging under the influence of gravity. Derive the equation for the period of a simple pendulum. Which of the following changes would affect the period  $T$  of the pendulum's oscillation?

1. Increasing the mass  $m$  of the pendulum bob.
2. Increasing the length  $L$  of the string.
3. Increasing the amplitude (maximum angle) of the swing.
4. Decreasing the acceleration due to gravity  $g$ .

*Working Space*

*Answer on Page 79*

This exercise showed you that the period of a simple pendulum depends on the length

of the pendulum and the acceleration due to gravity, but not on the mass of the bob or the amplitude of the swing (for small angles). One derivation from the exercise above was the Equation ??, which can be rearranged as  $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$ . Solving this differential equation<sup>2</sup> will yield the angular position function  $\theta(t) = \theta_{\max} \sin(\omega t + \phi_0)$  or equivalently,  $\theta(t) = \theta_{\max} \sin(\sqrt{\frac{g}{L}}t + \phi_0)$  of the pendulum over time, which describes its oscillatory motion. Note that this differential equation does not depend on mass of the bob nor the amplitude of the swing  $\theta_{\max}$ , a core factor of pendulum harmonic motion.

### Exercise 13 Pendulum on the Moon

A simple pendulum has a period of 2 seconds when measured on Earth. If the same pendulum were taken to the Moon, where the acceleration due to gravity is approximately  $\frac{1}{6}$ , what would be its new period?

*Working Space*

- a Approximately 0.8 seconds
- b Approximately 2 seconds
- c Approximately 4.9 seconds
- d Approximately 6.8 seconds

*Answer on Page 80*

## 5.6 Alternating Current

We have talked about Alternating Current (AC) in previous chapters, but now we will explore its connection to oscillations. AC is an electric current that periodically reverses direction, in contrast to Direct Current (DC), which flows in a single direction and provides constant voltage. The voltage and current in an AC circuit vary sinusoidally with time, making them ideal for modeling using oscillatory functions.

In Direct Current,  $V = V_0$ , which means the original voltage is equal to the voltage everywhere else.

In an AC circuit, the voltage at any time  $t$  behaves exactly like a mass on a spring, following

<sup>2</sup>We will go more in depth about this in a future chapter.



a sinusoidal oscillation:

$$V(t) = V_{\text{peak}} \sin(\omega t + \phi) \quad (5.16)$$

where

- $V_{\text{peak}}$  is the maximum voltage reached in either direction
- $\omega$  is the angular frequency, related to standard frequency by  $\omega = \frac{2\pi}{T} = 2\pi f$
- $\phi$  which is the phase constant, or the starting position of the wave

Note that, because of the trigonometric nature of sine, the wave fluctuates between  $V_{\text{peak}}$  and  $-V_{\text{peak}}$ .

For sinusoidal voltages, the root mean squared (RMS) is derived as

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T [V_{\text{peak}} \sin(\omega t + \phi)]^2 dt} \\ &= V_{\text{peak}} \sqrt{\frac{1}{2T} \int_0^T [1 - \cos(2\omega t + 2\phi)] dt} \\ &= V_{\text{peak}} \sqrt{\frac{1}{2T} \int_0^T dt} \\ &= \frac{V_{\text{peak}}}{\sqrt{2}} \end{aligned}$$

### 5.6.1 Impedance

In DC, we use Resistance ( $R$ ). In AC oscillations, we use *Impedance*. Impedance is the “total opposition” to the oscillation and is a combination of standard resistance and Reactance ( $X$ ).

Capacitors oppose changes in voltage, which we call Capacitive Reactance:

$$X_C = \frac{1}{\omega C}$$

Inductive Reactance comes from inductors which oppose changes in current (acting like “electrical inertia”).

$$X_L = \omega L$$

Similarly, in AC circuits, the voltage and current waves do not always peak at the same time.

- In a purely inductive circuit, the current falls behind the voltage by  $90^\circ$  ( $\pi/2$ ).
- In a purely capacitive circuit, the current leads the voltage by  $90^\circ$  ( $\pi/2$ ).

### 5.6.2 Comparing Mass-Springs and AC

| Mechanical System (Mass-Spring) | Electrical System (AC Circuit)   |
|---------------------------------|----------------------------------|
| Displacement ( $x$ )            | Charge ( $q$ )                   |
| Velocity ( $v$ )                | Current ( $i$ )                  |
| Mass ( $m$ )                    | Inductance ( $L$ )               |
| Spring Constant ( $k$ )         | Reciprocal Capacitance ( $1/C$ ) |
| Damping/Friction ( $b$ )        | Resistance ( $R$ )               |

## 5.7 Summary

In this chapter, we have discussed oscillations via Simple Harmonic Motion, mass-springs, spring energy, pendulums, and AC currents.

- **Simple Harmonic Motion (SHM):** A specific type of periodic motion where a restoring force (like a spring) constantly pulls an object back toward an equilibrium point. Key Parameters include:
  - Oscillation above and below (or to and from) an equilibrium point.
  - A restoring force, proportional to displacement.
  - An angular velocity that is always changing,  $\omega$
  - Time for one full cycle ( $T = 2\pi/\omega$ ).
- **Hooke's Law:** For an ideal spring,  $F_s = -kx$ . This linear relationship leads to an angular frequency of  $\omega = \sqrt{k/m}$ .
- **Differential Equations:** Oscillations can be represented in the form  $m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = 0$ , where  $m$  is the mass,  $c$  is the damping constant, and  $k$  is the spring constant.
  - Underdamped: Oscillates with decaying amplitude.
  - Critically Damped: Returns to equilibrium as quickly as possible without oscillating.
  - Overdamped: Returns to equilibrium slowly without oscillating.
- **Energy Dynamics:** In a closed system, energy oscillates between Kinetic ( $K$ ) and Potential ( $U$ ). Total mechanical energy remains constant ( $E = \frac{1}{2}kA^2$ ), with  $K$  peaking at equilibrium and  $U$  peaking at maximum displacement.

- **Acceleration:** Acceleration is affected by both *centripetal acceleration* and *tangential acceleration*.

This chapter leads us into our next chapter on orbits, the oscillations of planetary motion.



# Orbits

Gravity is the force that attracts two bodies toward each other. It is responsible for the behavior of orbital motion. Gravity is described by Newton's Law of Universal Gravitation, which states that every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The formula for gravitational force is:

$$F_g = \frac{Gm_1m_2}{r^2}$$

Where  $F_g$  is the gravitational force between two objects,  $G$  is the gravitational constant,  $m_1$  and  $m_2$  are the masses of the objects, and  $r$  is the distance between the centers of the two objects.  $G$  is a constant with a value of approximately  $6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ .

If we think about acceleration due to gravity where one mass is significantly larger than the other, we can rewrite the equation as:

$$A_g = \frac{Gm_1m_2}{r^2m_2} = \frac{Gm_1}{r^2}$$

Where  $m_1$  is the mass of the more massive object and  $m_2$  is the mass of the significantly less massive object. This cancelling of the  $m_2$  mass is why the acceleration due to gravity is independent of the mass of the object in free fall.

A satellite stays in orbit around the planet because the pull of the planet's gravity causes it to accelerate toward the center of the planet.

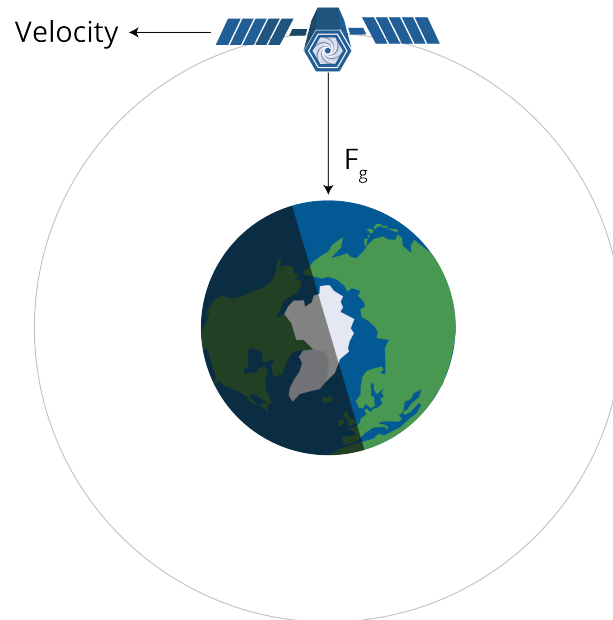


Figure 6.1: A satellite's centripetal force is gravity.

The satellite must be moving at a very particular speed to keep a constant distance from the planet — to travel in a circular orbit. If it is moving too slowly, it will get closer to the planet. If it is going too fast, it will get farther from the planet.

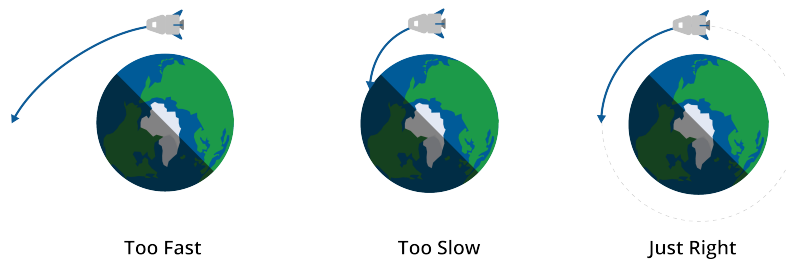


Figure 6.2: A diagram showing the required speeds for entering orbit.

The radius of a satellite in a low orbit is typically about 2 million meters above the ground. At that distance, the acceleration due to gravity is more like  $6.8\text{m/s}^2$ , instead of the  $9.8\text{m/s}^2$  that we experience on the surface of the planet.

How fast does the satellite need to be moving in a circle with a radius of 8.37 million meters to have an acceleration of  $6.8\text{m/s}^2$ ? Real fast.

Recall that the acceleration vector is

$$a = \frac{v^2}{r}$$

Thus the velocity  $v$  needs to be:

$$v = \sqrt{ar} = \sqrt{6.8(8.37 \times 10^6)} = 7,544 \text{ m/s}$$

(That's 16,875 miles per hour.)

When a satellite falls out of orbit, it enters the atmosphere at that 7,544 m/s. The air rushing by generates so much friction that the satellite gets very, very hot, and usually disintegrates.

## 6.1 Astronauts are not weightless

Some people see astronauts floating inside an orbiting spacecraft and think there is no gravity: that the astronauts are so far away that the gravity of the planet doesn't affect them. This is incorrect. The gravity might be slightly less (maybe 6 newtons per kg instead of 9.8 newtons per kg), but the weightlessness they experience is because they and the spacecraft are in free fall. They are moving so fast (in a direction perpendicular to gravity) that they don't collide with the planet.

**Exercise 14      Mars Orbit**

Working Space

The radius of Mars is 3.39 million meters. The atmosphere goes up another 11 km. Let's say you want to put a satellite in a circular orbit around Mars with a radius of 3.4 million meters.

The acceleration due to gravity on the surface of Mars is  $3.721\text{m/s}^2$ . We can safely assume that it is approximately the same 11 km above the surface.

How fast does the satellite need to be traveling in its orbit? How long will each orbit take?

Answer on Page 81

**6.2 Geosynchronous Orbits**

The planet Earth rotates once a day. Satellites in low orbits circle the Earth many times a day. Satellites in very high orbits circle less than once per day. There is a radius at which a satellite orbits exactly once per day. Satellites at this radius are known as "geosynchronous" or "geostationary", because they are always directly over a place on the planet.



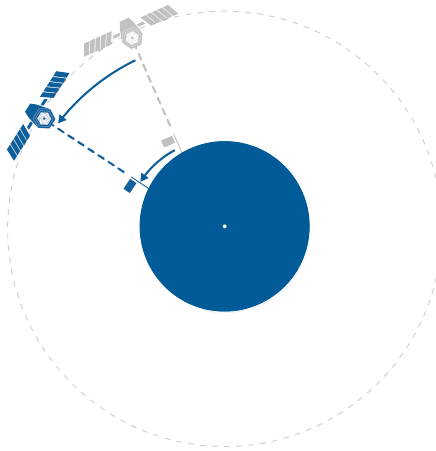


Figure 6.3: A satellite in geosynchronous orbit.

The radius of a circular geosynchronous orbit is 42.164 million meters. (About 36 km above the surface of the earth.)

A geosynchronous satellite travels at a speed of 3,070 m/s.

Geosynchronous satellites are used for the Global Positioning Satellite system, weather monitoring system, and communications system.

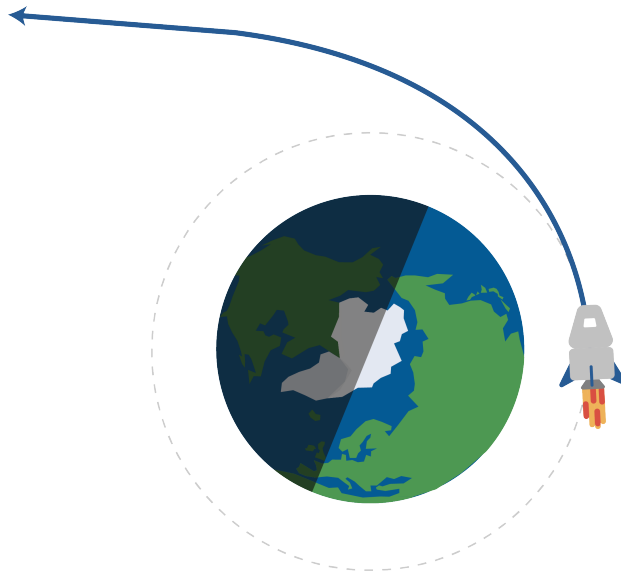


Figure 6.4: A satellite can reach a speed at which it "escapes" earth's orbit (centripetal force).

### 6.3 Kepler's Laws

There are three laws that describe the motion of planets and their elliptical properties.

1. Every planet moves along an ellipse with the sun at one focus.

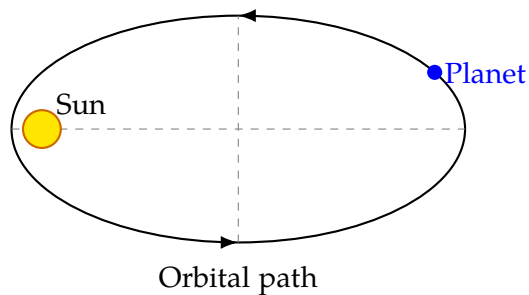


Figure 6.5: A diagram of Kepler's First Law.

2. As a planet moves around its orbit, it sweeps out equal areas in equal times. This means that a planet moves faster when it is closer to the sun and slower when it is farther from the sun. Note that this directly comes from the *Conservation of Angular Momentum*.

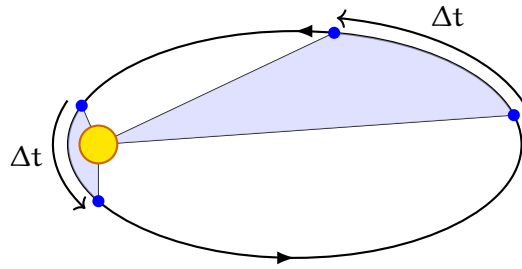


Figure 6.6: A diagram of Kepler's Second Law.

3. If  $T$  is the period of a planet's orbit, and  $a$  is the length of the semi-major axis of the ellipse, then  $\frac{T^2}{a^3}$  is the same for all planets orbiting the same star.

## 6.4 Escape velocity

The escape velocity is the minimum speed needed for an object to “break free” from the gravitational attraction of a massive body or planet, without further propulsion. Gravity will never pull it back, because we assume no further forces act on the object. This is most commonly associated with a spaceship or satellite escaping from a planet's gravitational pull.

We need to understand that escape velocity relies on energy, rather than a force or acceleration. An object must have enough kinetic energy to overcome the gravitational potential energy pulling it back toward the planet. To escape from an orbit, the object's mechanical energy (the sum of its kinetic and potential energy) must be greater than or equal to zero.

We can derive its initial energy as:  $E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{R}$ , where  $m$  is the mass of the object,  $M$  is the mass of the planet,  $R$  is the distance from the center of the planet to the object, and  $G$  is the gravitational constant.

After its escape, the object will be infinitely far away from the planet, and its potential energy will be zero. If we assume it just barely escapes, its kinetic energy will also be zero. Therefore, we set the initial energy equal to zero:

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0 \quad (6.1)$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \quad (6.2)$$

$$v^2 = \frac{2GM}{R} \quad (6.3)$$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} \quad (6.4)$$

Notice that the mass of the object  $m$  cancels out, meaning that the escape velocity is

independent of the mass of the escaping object. It only depends on the mass of the planet and the distance from its center.

Further, **notice that the escape velocity is  $\sqrt{2}$  times the orbital velocity**,  $v_{\text{orb}} = \sqrt{GM/R}$  at the same distance from the planet's center. This means that to escape from a circular orbit, an object must increase its speed by about 41.4%. For Earth, the escape velocity at the surface is approximately 11.2 km/s (about 25,000 mph). This means that any object, regardless of its mass, must reach this speed to escape Earth's gravitational pull without further propulsion.

### Exercise 15      Escape Velocity from Mars

*Working Space*

The radius of Mars is  $3.4 \times 10^6$  meters.  
The mass of Mars is  $6.42 \times 10^{23}$  kg. A  
spaceship is sitting on the surface of Mars.

1. Find the escape velocity from the surface of Mars.
2. Suppose the spaceship is in a circular orbit 11 km above the surface of Mars. What is the escape velocity from that orbit?

*Answer on Page 81*

Note that giving an object an extra height  $h$  causes the escape velocity to decrease (as escape velocity is inversely proportional to the square root of the radius plus height). This is due to gravitation potential energy being less negative at larger radii. However, the effect is small compared to the overall radius of the planet. Problems may give you a radius plus height, or just the radius of the planet, or a total distance from the center, depending on the context, so read each problem carefully.

# Polar Coordinates

We have already seen how to plot a function with  $(x, y)$  coordinates. For every  $x$  that we put into a function, it returns a  $y$ . These pairs of coordinates tell us where on the  $xy$ -plane to graph the function. This coordinate system, where  $x$  and  $y$  are oriented horizontally and vertically, is called the *Cartesian* coordinate system. It can be used to describe 2D space, but it is not the only way.

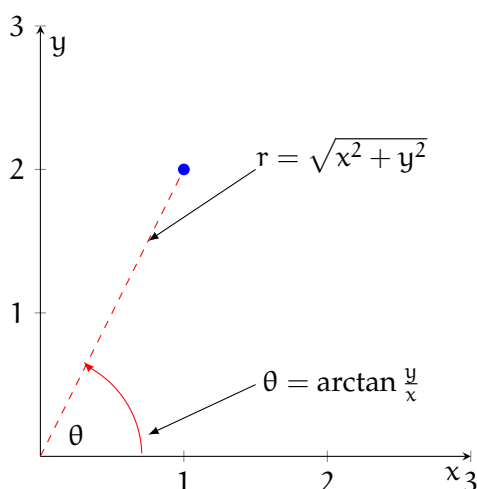


Figure 7.1: The point  $(1, 2)$  is  $\sqrt{5}$  units from the origin and approximately 1.107 radians counterclockwise from horizontal

Instead of thinking about the horizontal and vertical position, we could think about distance from the origin and rotation about the origin. Take the Cartesian coordinate point  $(1, 2)$  (see figure 7.1). How far is  $(1, 2)$  from the origin,  $(0, 0)$ ? We can create a right triangle, where the legs are parallel to the  $x$  and  $y$  axes. This means the leg lengths are 1 and 2, and we can use the Pythagorean theorem to find the length of the hypotenuse (which is the distance from the origin to the point):

$$c^2 = a^2 + b^2$$

$$c^2 = 1^2 + 2^2 = 1 + 4 = 5$$

$$c = \sqrt{5}$$

Therefore, the Cartesian point  $(1, 2)$  is  $\sqrt{5}$  units from the origin. This is not enough to find our point: there are infinite points that are  $\sqrt{5}$  from the origin (see 7.2). To identify a

particular point that is a distance of  $\sqrt{5}$  from the origin, we also need an *angle of rotation*. By convention, angles are measured from the positive  $x$ -axis. This means points on the positive  $x$ -axis have an angle of  $\theta = 0$ , points on the positive  $y$ -axis have an angle of  $\theta = \frac{\pi}{2}$ , and so on.

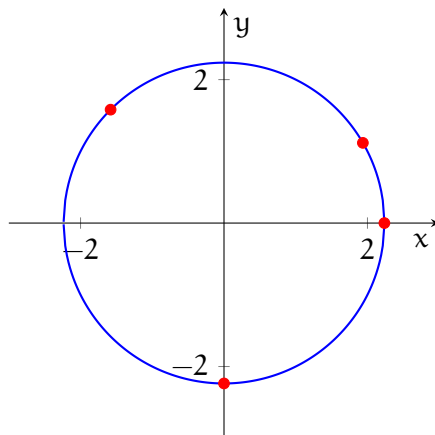


Figure 7.2: There are infinite points  $\sqrt{5}$  from the origin, represented by the circle with a radius of  $\sqrt{5}$  centered about the origin

We can use trigonometry to find the appropriate angle of rotation for our Cartesian point. There are many ways to do this, but using  $\arctan$  is the most straightforward. Recall that:

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

That is, for a given angle in a right triangle, the tangent of that angle is given by the length of the opposite leg divided by the adjacent leg. In our case, the opposite leg is the vertical distance ( $y$ -value of the Cartesian point) and the adjacent leg is the horizontal distance ( $x$ -value of the Cartesian point), which means:

$$\tan \theta = \frac{2}{1}$$

$$\theta = \arctan 2 \approx 1.107 \text{ radians}$$

## 7.1 Plotting Polar Coordinate Points

How do we plot polar coordinate points? Begin by locating the angle given by the second coordinate (remember, the angle is measured counterclockwise from the horizontal). Your point will lie somewhere on this line. Next, move outwards along the angle by the radius given by the first coordinate.

**Example:** Plot the polar coordinate point  $(2, \frac{\pi}{3})$ .

**Solution:** Begin by locating  $\theta = \frac{\pi}{3}$  (see figure 7.3)

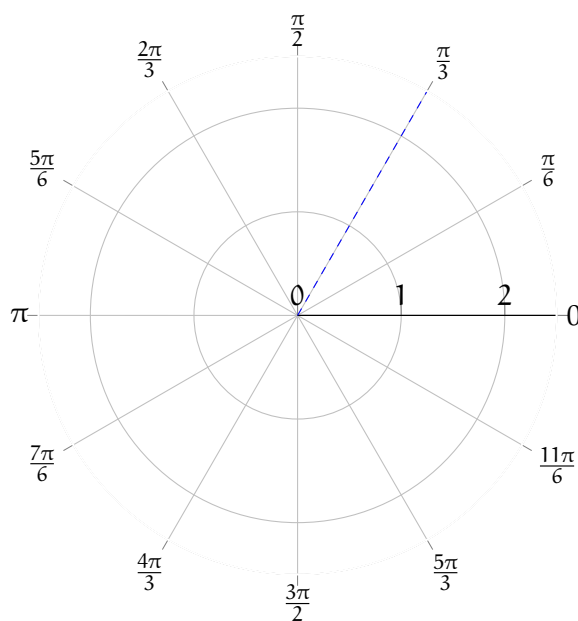


Figure 7.3:  $\theta = \frac{\pi}{3}$

Then, move your finger or pencil along the line  $\theta = \frac{\pi}{3}$  until you reach  $r = 2$  (see figure 7.4).

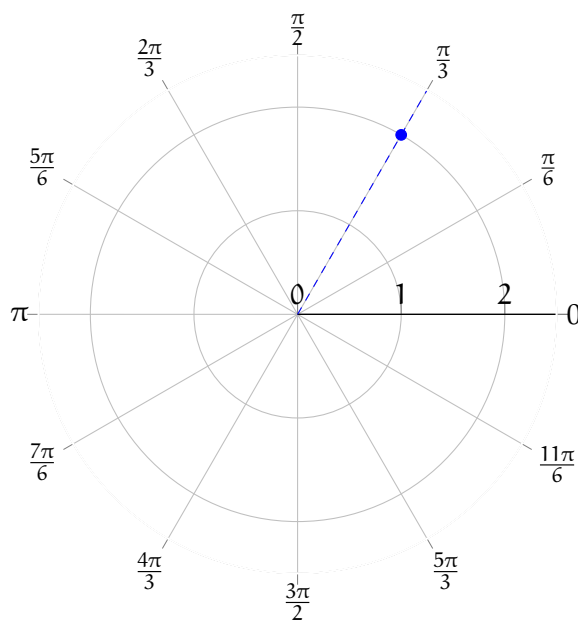


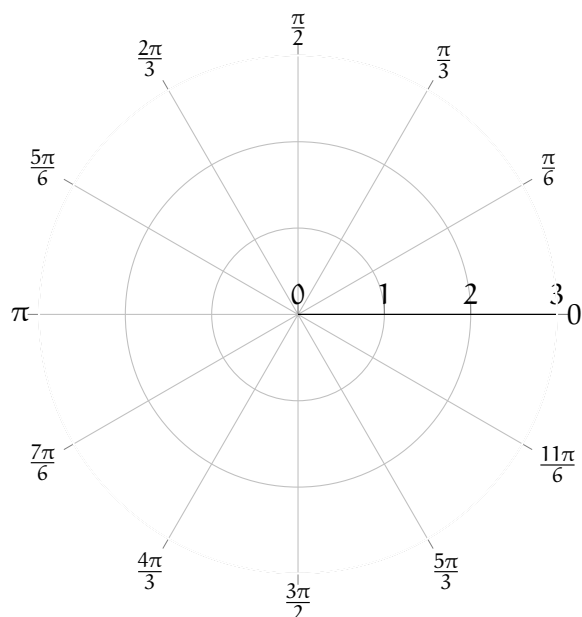
Figure 7.4:  $(2, \frac{\pi}{3})$

**Exercise 16**

Plot the following polar coordinate points on the provided polar axis (hint: negative angles are taken counterclockwise):

1.  $(1, \pi)$
2.  $(1.5, \frac{\pi}{2})$
3.  $(1.5, -\frac{\pi}{6})$
4.  $(2, \frac{3\pi}{4})$

*Working Space*



*Answer on Page 82*

**7.2 Equivalent Points**

Unlike the Cartesian coordinate system, two different coordinates may lie at the same location. Consider the points  $(1, \frac{\pi}{4})$  and  $(-1, \frac{5\pi}{4})$  (see figure 7.5). When a radius is negative, you move *backwards* back over the origin, like a mirror image.



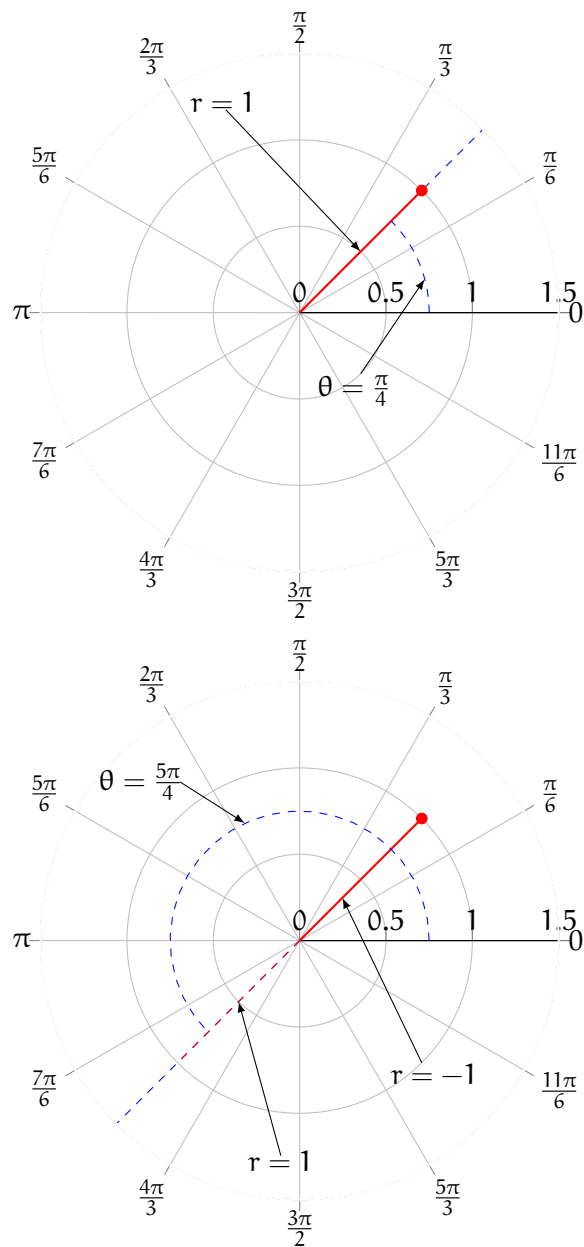


Figure 7.5: The polar coordinates points  $(1, \frac{\pi}{4})$  and  $(-1, \frac{5\pi}{4})$  are the same location on a polar axis

## 7.3 Changing coordinate systems

### 7.3.1 Cartesian to Polar

From the example above, you should see that a given Cartesian coordinate,  $(x, y)$ , can also be expressed as a polar coordinate,  $(r, \theta)$ , where  $r$  is the distance from the origin and  $\theta$  is the angle of rotation from the horizontal. (Note: Polar functions are generally given as  $r$  defined in terms of  $\theta$ , which means the *dependent* variable is listed first in the coordinate pair, unlike Cartesian coordinates.) Additionally,

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan \frac{y}{x}$$

**Example:** Express the Cartesian point  $(-3, 4)$  in polar coordinates.

**Solution:** Taking  $x = -3$  and  $y = 4$ , we find that:

$$r = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

We follow the convention of only taking the positive solution to the square root. Finding  $\theta$ :

$$\theta = \arctan \frac{4}{-3}$$

When you evaluate the arctan with a calculator, you are likely to get back  $\theta = -0.928$ . Recall that  $\tan \theta = \tan \theta \pm n\pi$ , where  $n$  is an integer. We know our Cartesian point,  $(-3, 4)$ , is in the II quadrant, while the angle  $-0.928$  radians would fall in the IV quadrant. So, clearly,  $-0.928$  radians is not correct. Most calculators restrict the output of arctan to angles between  $-\frac{\pi}{2}$  and  $\frac{\pi}{2}$ , because there are actually multiple angles where  $\tan \theta = -\frac{4}{3}$ . Since  $\tan \theta = \tan \theta \pm n\pi$ , we also know that:

$$\arctan -\frac{4}{3} = -0.928 \pm n\pi$$

Another possible  $\theta$  is  $-0.928 + \pi \approx 2.214$ , which does fall in the appropriate quadrant. This means the polar coordinates  $(5, 2.214)$  are the same as the Cartesian coordinates  $(-3, 4)$ . *Note:* It is standard practice to express angles in radians, and not degrees, when using polar coordinates.

### 7.3.2 Polar to Cartesian

We can also leverage our knowledge of right triangles to convert polar coordinates to Cartesian coordinates. Take the polar coordinate  $(2, \frac{\pi}{4})$  (see figure 7.6). We can draw a right triangle with legs parallel to the  $x$  and  $y$  axes (not shown in the figure) and a hypotenuse that goes from the origin to the polar coordinate  $(2, \frac{\pi}{4})$ .

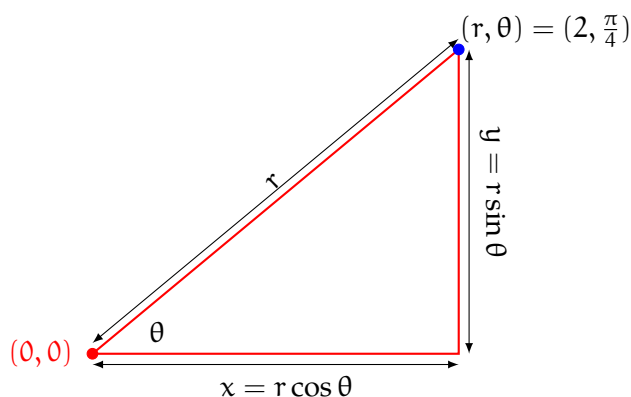


Figure 7.6: To convert from polar to Cartesian coordinates, use the identities  $x = r \cos \theta$  and  $y = r \sin \theta$

Recall from trigonometry that:

$$\sin \theta = \frac{\text{opposite leg}}{\text{hypotenuse}}$$

We know that the hypotenuse of this triangle has a length of  $r$ . The opposite leg is vertical and is the same length as the distance of the polar coordinate from the  $x$ -axis. Therefore, the length of the vertical leg represents the  $y$  value of that same polar coordinate if it were expressed in Cartesian coordinates. So, we can say that:

$$\sin \theta = \frac{y}{r}$$

And therefore:

$$y = r \sin \theta$$

By a similar process, we also see that:

$$x = r \cos \theta$$

This is easy to visualize and understand for  $0 \leq \theta \leq \frac{\pi}{2}$ , but it also holds for other values of  $\theta$ .

**Example:** Express the polar coordinate  $(\frac{3}{2}, \frac{2\pi}{3})$  in Cartesian coordinates.

**Solution:** From the polar coordinate, we see that  $\theta = \frac{2\pi}{3}$  and  $r = \frac{3}{2}$ . Therefore:

$$x = r \cos \theta = \frac{3}{2} \cdot \cos \frac{2\pi}{3} = \frac{3}{2} \cdot -\frac{1}{2} = -\frac{3}{4}$$

$$y = r \sin \theta = \frac{3}{2} \cdot \sin \frac{2\pi}{3} = \frac{3}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$$

The Cartesian coordinate  $(-\frac{3}{4}, \frac{3\sqrt{3}}{4})$  has the same location as the given polar coordinate.

### Exercise 17

Convert the following polar coordinates to Cartesian coordinates:

*Working Space*

1.  $(2, \frac{3\pi}{2})$
2.  $(\sqrt{2}, \frac{3\pi}{4})$
3.  $(3, -\frac{\pi}{4})$
4.  $(-3, -\frac{\pi}{3})$
5.  $(2, -\frac{\pi}{2})$

*Answer on Page 82*

**Exercise 18**

Convert the following Cartesian coordinates to polar coordinates. Restrict  $\theta$  to  $0 \leq \theta < 2\pi$ .

1.  $(-4, 4)$
2.  $(3, 3\sqrt{3})$
3.  $(\sqrt{3}, -1)$
4.  $(-6, 0)$
5.  $(-2, -2)$

*Working Space*

*Answer on Page 82*

**7.4 Circles in Polar Coordinates**

Many conic sections, including circles, are simpler to express as polar functions than as Cartesian functions. Consider a circle with a radius of 2 centered about the origin. The polar function for this is  $r = 2$  for all  $\theta$ . Let's write a Cartesian function for the same circle.

We know that for every point on the circle, the distance to the origin is 2. This means that, by the Pythagorean theorem,

$$r^2 = x^2 + y^2$$

.

(see figure 7.7)

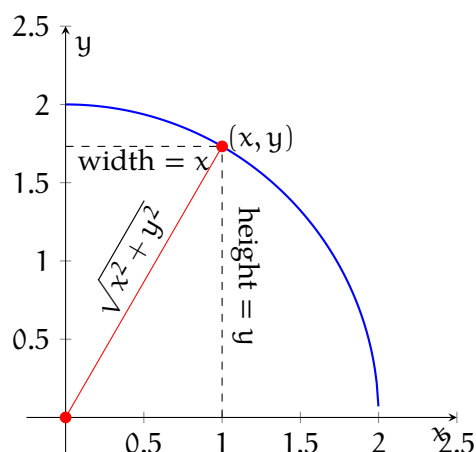


Figure 7.7: All  $(x, y)$  pairs on the circle are the same distance from the origin

We can solve this equation for  $y$ , given that  $r = 2$  (in this case):

$$y = \pm \sqrt{2^2 - x^2}$$

Notice that this is really two equations:  $y = \sqrt{2^2 - x^2}$  and  $y = -\sqrt{2^2 - x^2}$ . This is more complex than the polar equation,  $r = 2$ .

As seen above, the equation of a circle with radius  $R$  centered on the origin is simply  $r = R$  in polar coordinates. What if we want a circle centered somewhere else? Polar coordinates are best when a circle is bisected by the  $x$  or  $y$  axis. Consider the polar equation  $r = 3 \sin \theta$ . Let's use a table to find some points and plot the function:

| $\theta$         | $r = 3 \sin \theta$   |
|------------------|-----------------------|
| 0                | 0                     |
| $\frac{\pi}{6}$  | $\frac{3}{2}$         |
| $\frac{\pi}{4}$  | $\frac{3\sqrt{2}}{2}$ |
| $\frac{\pi}{3}$  | $\frac{3\sqrt{3}}{2}$ |
| $\frac{\pi}{2}$  | 3                     |
| $\frac{2\pi}{3}$ | $\frac{3\sqrt{3}}{2}$ |
| $\frac{3\pi}{4}$ | $\frac{3\sqrt{2}}{2}$ |
| $\frac{5\pi}{6}$ | $\frac{3}{2}$         |
| $\pi$            | 0                     |

Here is how those points look plotted (see figures 7.8 and 7.9):

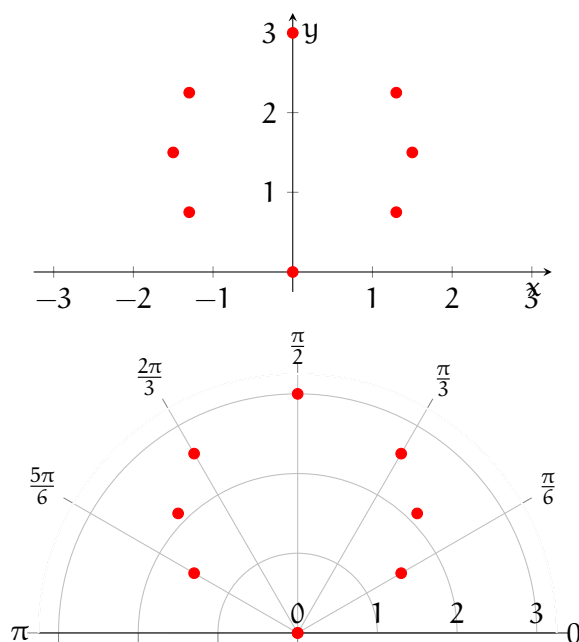


Figure 7.8: Several points for  $r = 3 \sin \theta$  plotted on Cartesian and polar coordinate systems

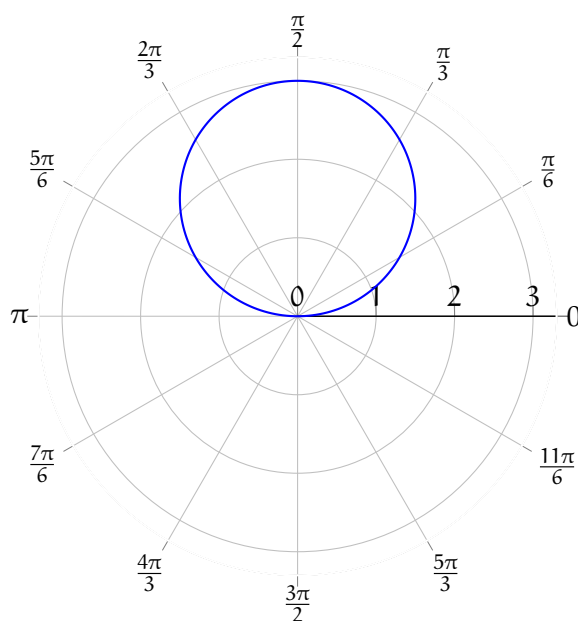
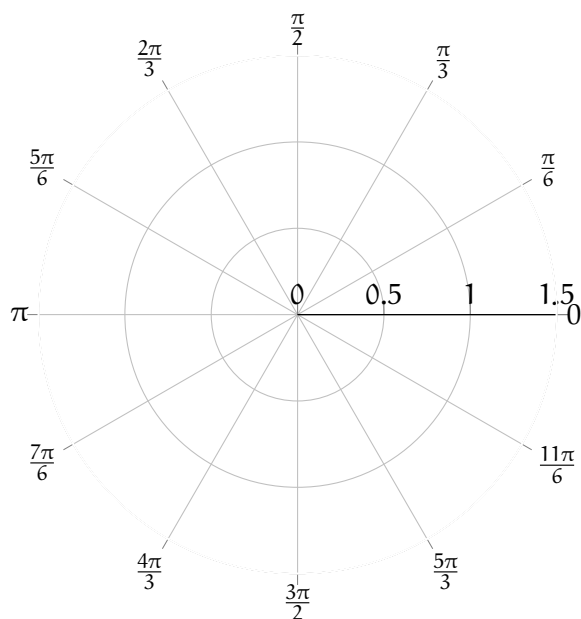


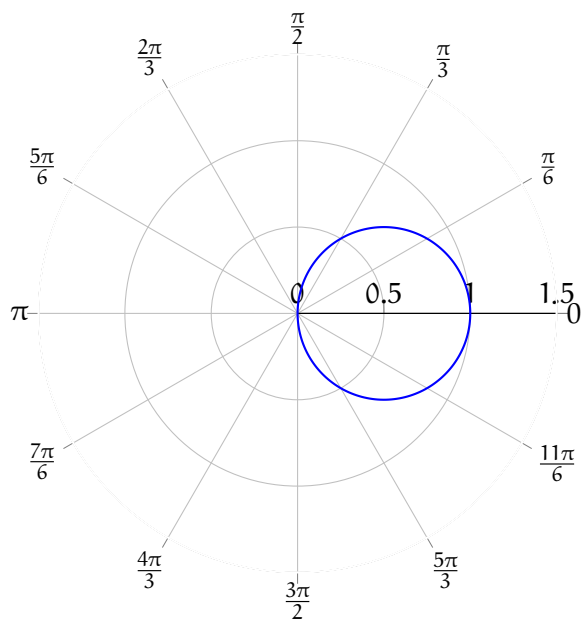
Figure 7.9:  $r = 3 \sin \theta$  plotted on a polar coordinate system

So, the polar equation  $r = 3 \sin \theta$  gives a circle with radius  $\frac{3}{2}$  centered at  $(0, \frac{3}{2})$ .

**Example:** Describe the graph of  $r = \cos \theta$ . Feel free to make a rough plot on the blank polar axis below:



**Solution:** This plot will look like a circle of radius 0.5 centered at  $(0.5, 0)$  (in polar coordinates).



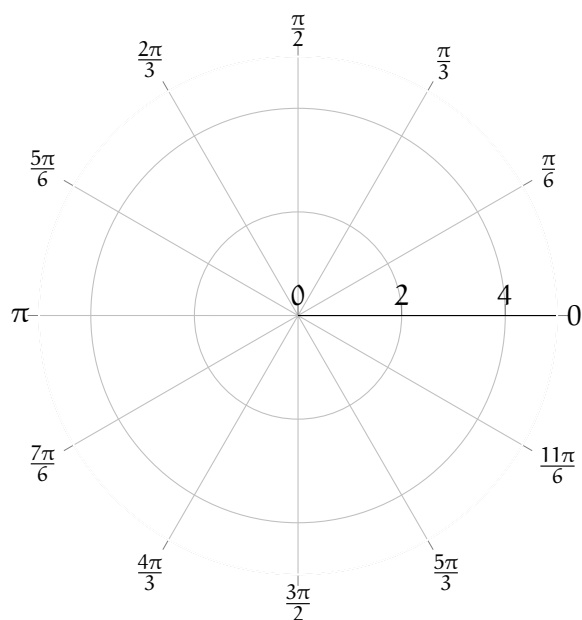


### Exercise 19

Sketch the following polar functions on the provided polar axis for  $0 \leq \theta < 2\pi$ :

1.  $r = 3$
2.  $\theta = \pi$
3.  $r = 2 \cos \frac{\theta}{2}$
4.  $r = -4 \sin \theta$
5.  $r = \theta$

Working Space



Answer on Page 83



# Answers to Exercises

### Answer to Exercise 1 (on page 3)

$$\frac{300 \times 10^6}{5.66 \times 10^{14}} = 530 \times 10^{-9} = 530 \text{ nm}$$

### Answer to Exercise 2 (on page 14)

Assuming the mirror is truly vertical and the floor is truly horizontal, the new cut off should be exactly the same as the old one: It should be below your chin the same amount that your eyes are above your chin.

*Illustration Here*

### Answer to Exercise 3 (on page 15)

*Are there white photons?* No. What we call “white” is a blend of photons that are several different colors.

Some people like to say white light is the combination of all visible colors of photons in equal amounts. That seems oddly specific and unusual.

Maybe it is better to imagine it from the human experience of white light. In our eyes, we have three different types of color-sensing cones, which generally correspond to the red, green, and blue regions of the spectrum. When all three are excited about equal amounts, humans experience that as white. On your computer screen, for example, what you see as white is just a blend of three colors of photons: a red, a green, and a blue.

*Are there black photons?* No. What we call “black” is an absence of photons in the visible range.

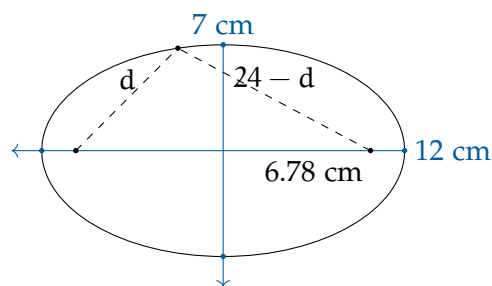
*Are there yellow photons?* Yes! There is a region of the color spectrum that is yellow. It has a wavelength of about 527 nm. Photons at this energy level excite both our green-sensitive and red-sensitive cones (we will expand on this in the eye chapter!). Your computer

monitor does not actually create light with a 527 nm wavelength. Instead, it creates red light and green light, which our eyes interpret as yellow.

### Answer to Exercise 4 (on page 18)

The length of the string is easy:  $2 \times 12 = 24$  cm.

The distance from the center to the focal point is  $\sqrt{12^2 - 7^2} \approx 6.78$  cm.



### Answer to Exercise 5 (on page 22)

We need the distance from the center out to each of the three axes. We know that  $a = \left(\frac{1}{2}\right) 30 = 15$  cm.

We can calculate the  $b$  and  $c$  (which are equal) using the circumference given:  $2b\pi = 60$ , so  $c = b \approx 9.55$  cm.

The volume, then is

$$V = \frac{4}{3}\pi(15)(9.55)(9.55) \approx 5,730 \text{ cubic centimeters}$$

The mass would be  $5,730 \times 11.34 = 64,973$  grams or about 65 kg.

### Answer to Exercise 6 (on page 38)

The correct answer is: The frequency would increase by a factor of  $\sqrt{2}$ .

The frequency of oscillation for a mass-spring system is given by the formula:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

where  $k$  is the spring constant and  $m$  is the mass attached to the spring. Since frequency is inversely proportional to the square root of the mass, if the mass is halved, the new frequency  $f'$  can be calculated as follows:

$$\begin{aligned} f' &= \frac{1}{2\pi} \sqrt{\frac{k}{m/2}} \\ &= \frac{1}{2\pi} \sqrt{\frac{2k}{m}} \\ &= \sqrt{2} \cdot \frac{1}{2\pi} \sqrt{\frac{k}{m}} \\ &= \sqrt{2} \cdot f \end{aligned}$$

Therefore, the frequency increases by a factor of  $\sqrt{2}$  when the

### Answer to Exercise 7 (on page 40)

First, we find the angular frequency  $\omega$  using the formula:

$$\omega = 2\pi f = \sqrt{\frac{k}{m}} = \sqrt{\frac{400 \text{ N/m}}{4 \text{ kg}}} = \sqrt{100} = 10 \text{ rad/s}$$

Therefore, the general equation for the position of a mass-spring system undergoing simple harmonic motion is:

$$x(t) = A \cos(\omega t + \phi)$$

becomes  $x = 6 \sin(10t + \phi)$  since the initial velocity is 0, meaning it starts at maximum displacement. Solving for  $\phi$  using the initial condition  $x(0) = 6 \text{ cm}$ :

$$6 = 6 \sin(\phi) \implies \sin(\phi) = 1 \implies \phi = \frac{\pi}{2}$$

So the base equation becomes:

$$x(t) = 6 \sin(10t + \frac{\pi}{2})$$

Which is option C.

### Answer to Exercise 8 (on page 41)

To find the speed of the ball as it leaves the spring, we can use the principle of conservation of mechanical energy. The net energy in the system remains constant, meaning that the potential energy stored in the compressed spring is converted into the kinetic energy of the ball when it is released. The potential energy stored in the spring when compressed is given by:

$$U_s = \frac{1}{2}kx^2 = \frac{1}{2}(500)(-0.24)^2 = 14.4 \text{ J}$$

The kinetic energy of the ball as it leaves the spring is the same amount, so we set the potential energy equal to the kinetic energy:

$$\frac{1}{2}mv^2 = U_s$$

Solving for  $v$ :

$$v = \sqrt{\frac{2U_s}{m}} = \sqrt{\frac{2(14.4)}{0.2}} = \sqrt{144} = 12 \text{ m/s}$$

The speed of the ball as it leaves the spring is 12 m/s.

### Answer to Exercise 9 (on page 43)

The correct differential equation is  $x'' + 100x = 0$ . This is derived from the equation of motion for any simple harmonic oscillator:  $mx'' + kx = 0$ , where  $m = 2 \text{ kg}$  and  $k = 200 \text{ N/m}$ . Dividing by  $m$ , we get  $x'' + \frac{k}{m}x = 0$ , which simplifies to  $x'' + 100x = 0$ .

### Answer to Exercise 10 (on page 44)

- This system is damped. Specifically, it is underdamped because the amplitude of oscillation decreases over time.
- To make this system undergo simple harmonic motion, we would need to remove the damping term from the equation of motion. This means setting the damping coefficient  $c$  to zero in the differential equation, which would eliminate the exponential decay factor in the position function. The equation becomes  $x(t) = 0.15e^{-0.40t} \cos(4t + 0.30) \rightarrow x(t) = 0.15 \cos(4t + 0.30)$

### Answer to Exercise 11 (on page 45)

1. Notice that the system equation is linear as it is defined with an  $x''$  and  $x$ . The general solution to the differential equation  $x'' + 16x = 0$  is of the form  $x(t) = A \cos(4t) + B \sin(4t)$ . This comes from the idea that the characteristic equation  $r^2 + 16 = 0$  has roots  $r = \pm 4i$ , where the  $\beta$  term is 4. Using the initial conditions  $x(0) = 0.20$  and  $v(0) = x'(0) = 0$ , we find our velocity equation to be  $v(t) = x'(t) = -4A \sin(4t) + 4B \cos(4t)$ .

From  $x(0) = A \cos 0 + B \sin 0 = A(1) + B(0) = A = 0.20$ , we find that  $A = 0.20$ .

From  $v(0) = -4A \sin 0 + 4B \cos 0 = -4(0.20)(0) + 4B(1) = 4B = 0$ , we find that  $B = 0$ .

Therefore, the solution is:

$$x(t) = 0.20 \cos(4t)$$

2. To find the acceleration, we double differentiate  $x(t)$  with respect to time: To find the velocity, we differentiate  $x(t)$  with respect to time:

$$v(t) = x'(t) = -0.80 \sin(4t)$$

$$a(t) = v'(t) = -3.20 \cos(4t)$$

Since cosine and sine functions oscillate between -1 and 1 (regardless of angular frequency  $\omega = 4$ ), the maximum velocity is 0.80 m/s and the maximum acceleration is 3.20 m/s<sup>2</sup>.

## Answer to Exercise 12 (on page 47)

We know that the restoring force for a pendulum is given by  $F = -mg \sin \theta$ . For small angles, we can approximate  $\sin \theta \approx \theta$  (in radians), leading to  $F \approx -mg\theta$ . For any point on the circle, we can relate the angular displacement  $\theta$  to the arc length  $x$  using  $x = L\theta$ , where  $L$  is the length of the pendulum. Thus, we can rewrite the restoring force in terms of the second derivative, which represents angular acceleration:

$$a_{\text{tan}} = \frac{d^2x}{dt^2} = L \frac{d^2\theta}{dt^2} \quad (5.9)$$

Using Newton's Second Law, we have:

$$mL \frac{d^2\theta}{dt^2} = -mg \sin \theta \quad (5.10)$$

$$L \frac{d^2\theta}{dt^2} + g \sin \theta = 0 \quad (5.11)$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0 \quad (5.12)$$

But for small angles, we can approximate  $\sin \theta \approx \theta$ , leading to:

$$[\text{label} = \text{eq : linearpendulum}] \frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \quad (5.13)$$

This is a second-order linear differential equation with constant coefficients, similar to the mass-spring system problems discussed earlier. The angular frequency  $\omega$  of the pendulum can be identified as:

$$\omega = \sqrt{\frac{g}{L}} \quad (5.14)$$

Therefore, the period  $T$  of the pendulum is given by:

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (5.15)$$

### Answer to Exercise 13 (on page 48)

The correct answer is: Approximately 4.4 seconds.

The period  $T$  of a simple pendulum is given by the formula:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where  $L$  is the length of the pendulum and  $g$  is the acceleration due to gravity.

On Earth, the period is 2 seconds, so we can set up the equation:

$$2 = 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}}$$

This shows that  $T$  is inversely proportional to the square root of  $g$ . We can rearrange this equation to solve for  $L$ . So if  $g$  decreases by a factor of 6, then  $T$  will increase by a factor of  $\sqrt{6}$ .

Now, on the Moon, the acceleration due to gravity is approximately  $\frac{1}{6}g_{\text{Earth}}$ . We can substitute this into the period formula:

$$T_{\text{Moon}} = 2\pi\sqrt{\frac{L}{g_{\text{Moon}}}} = 2\pi\sqrt{\frac{L}{\frac{1}{6}g_{\text{Earth}}}} = 2\pi\sqrt{\frac{6L}{g_{\text{Earth}}}} = \sqrt{6} \cdot 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}} = \sqrt{6} \cdot T_{\text{Earth}}$$



Since our original period on Earth is 2 seconds, we have:

$$T_{\text{Moon}} = \sqrt{6} \cdot 2 \approx 4.9 \text{ seconds}$$

Which is equal to option (c).

### Answer to Exercise ?? (on page 56)

$$v = \sqrt{3.721(3.4 \times 10^6)} = 3,557 \text{ m/s}$$

The circular orbit is  $2\pi(3.4 \times 10^6) = 21.4 \times 10^6$  meters in circumference.

The period of the orbit is  $(21.4 \times 10^6)/3,557 \approx 6,000$  seconds.

### Answer to Exercise 15 (on page 60)

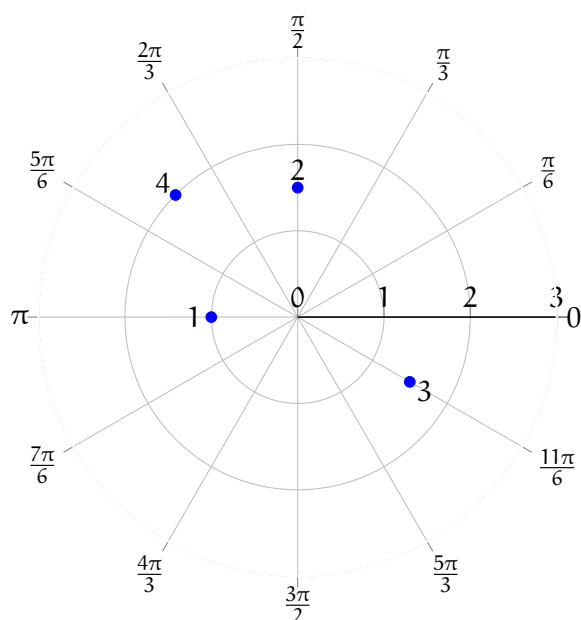
Using equation ??:

1.

$$\begin{aligned} v_{\text{esc}} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2(6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2)(6.42 \times 10^{23} \text{ kg})}{3.4 \times 10^6 \text{ m}}} \\ &\approx 5,027 \text{ m/s} \end{aligned}$$

2. Now the radius is  $3.4 \times 10^6 + 11,000 = 3.411 \times 10^6$  meters, because the satellite is 11 km above the surface. Recalculating:

$$\begin{aligned} v_{\text{esc}} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2(6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2)(6.42 \times 10^{23} \text{ kg})}{3.411 \times 10^6 \text{ m}}} \\ &\approx 5,013 \text{ m/s} \end{aligned}$$

**Answer to Exercise 16 (on page 64)****Answer to Exercise 17 (on page 68)**

1.  $(0, -2)$ .  $x = 2 \cdot \cos \frac{3\pi}{2} = 2 \cdot 0 = 0$  and  $y = 2 \cdot \sin \frac{3\pi}{2} = 2 \cdot -1 = -2$ .
2.  $(-1, 1)$ .  $x = \sqrt{2} \cdot \cos \frac{3\pi}{4} = \sqrt{2} \cdot -\frac{\sqrt{2}}{2} = -1$  and  $y = \sqrt{2} \cdot \sin \frac{3\pi}{4} = \sqrt{2} \cdot \frac{\sqrt{2}}{2} = 1$ .
3.  $(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2})$ .  $x = 3 \cdot \cos -\frac{\pi}{4} = 3 \cdot \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}$  and  $y = 3 \cdot \sin -\frac{\pi}{4} = 3 \cdot -\frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2}$ .
4.  $(-\frac{3}{2}, -\frac{3\sqrt{3}}{2})$ .  $x = (-3) \cdot \cos \frac{\pi}{3} = (-3) \cdot \frac{1}{2} = -\frac{3}{2}$  and  $y = (-3) \cdot \sin \frac{\pi}{3} = (-3) \cdot \frac{\sqrt{3}}{2} = -\frac{3\sqrt{3}}{2}$ .
5.  $(0, -2)$ .  $x = 2 \cdot \cos -\frac{\pi}{2} = 2 \cdot 0 = 0$  and  $y = 2 \cdot \sin -\frac{\pi}{2} = 2 \cdot -1 = -2$ .

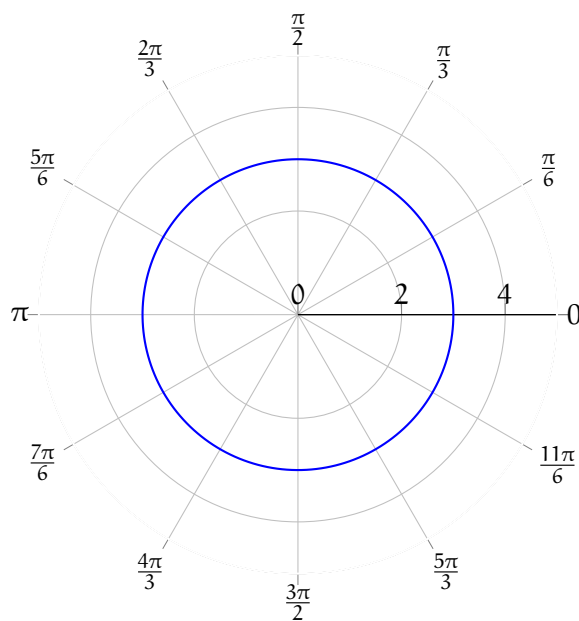
**Answer to Exercise 18 (on page 69)**

1.  $(4\sqrt{2}, \frac{3\pi}{4})$ .  $r = \sqrt{x^2 + y^2} = \sqrt{32} = 4\sqrt{2}$ .  $\arctan \frac{y}{x} = \arctan \frac{4}{-4} = \arctan -1 = -\frac{\pi}{4} + n\pi$ .  
We take  $\theta = \frac{3\pi}{4}$  to satisfy the domain restriction and be in the correct quadrant.
2.  $(6, \frac{\pi}{3})$ .  $r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$ .  $\arctan \frac{3\sqrt{3}}{3} = \arctan \sqrt{3} = \frac{\pi}{3} + n\pi$ .  
We take  $\theta = \frac{\pi}{3}$  to satisfy the domain restriction and be in the correct quadrant.

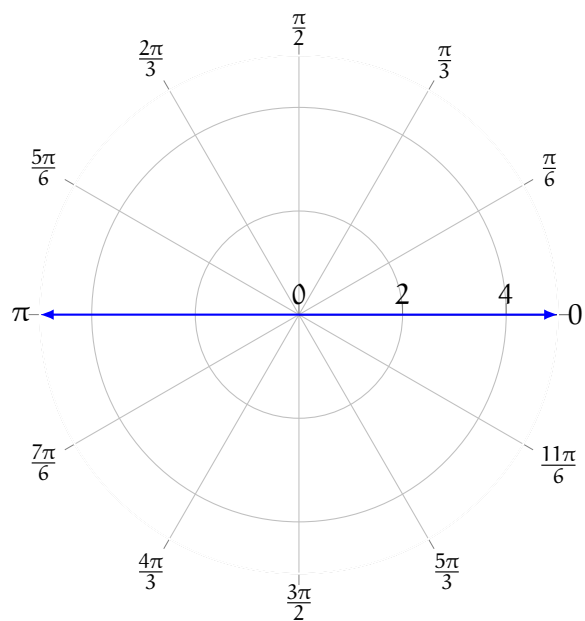
3.  $(2, \frac{11\pi}{6})$ .  $r = \sqrt{\sqrt{3}^2 + (-1)^2} = \sqrt{3+1} = 2$ .  $\arctan \frac{-1}{\sqrt{3}} = -\frac{\pi}{6} + n\pi$ . We take  $\theta = \frac{11\pi}{6}$  to satisfy the domain restriction and have the point in the correct quadrant.
4.  $(6, \pi)$ .  $r = \sqrt{(-6)^2 + 0^2} = 6$ .  $\arctan \frac{0}{-6} = \pi + n\pi$ . We take  $\theta = \pi$  to satisfy the domain restriction.
5.  $(2\sqrt{2}, \frac{5\pi}{4})$ .  $r = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$ .  $\arctan \frac{-2}{-2} = \arctan 1 = \frac{\pi}{4} + n\pi$ . We take  $\theta = \frac{5\pi}{4}$  to satisfy the domain restriction and be in the correct quadrant.

### Answer to Exercise ?? (on page 73)

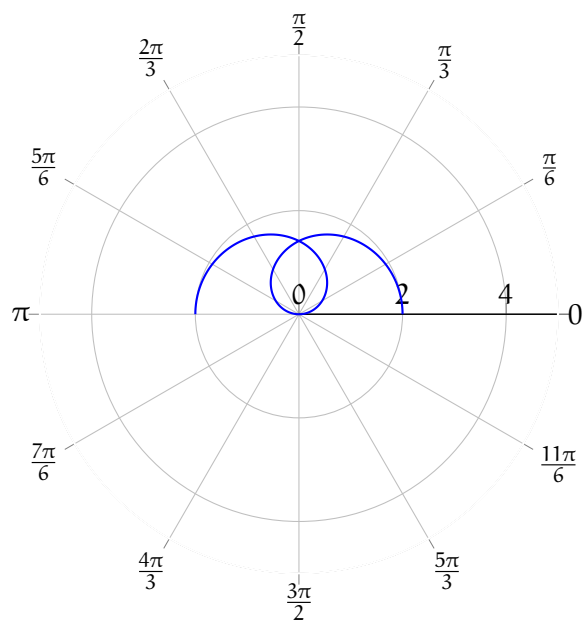
1.  $r = 3$



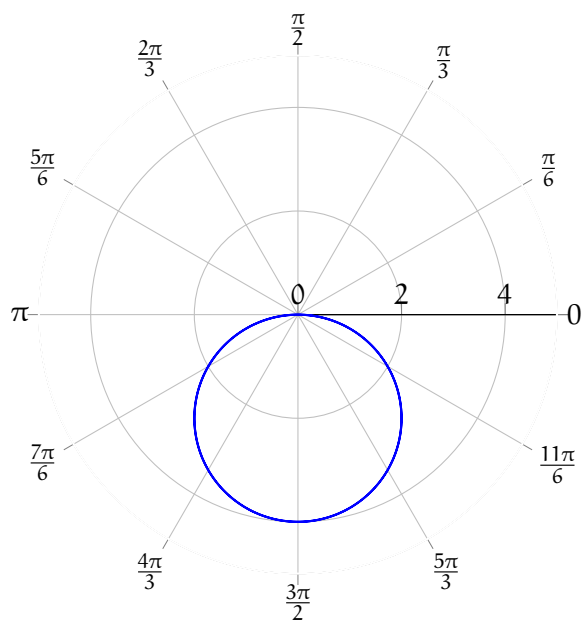
2.  $\theta = \pi$  Because  $r$  includes all real numbers, negative  $r$  is possible and the line  $\theta = \pi$  extends in both directions



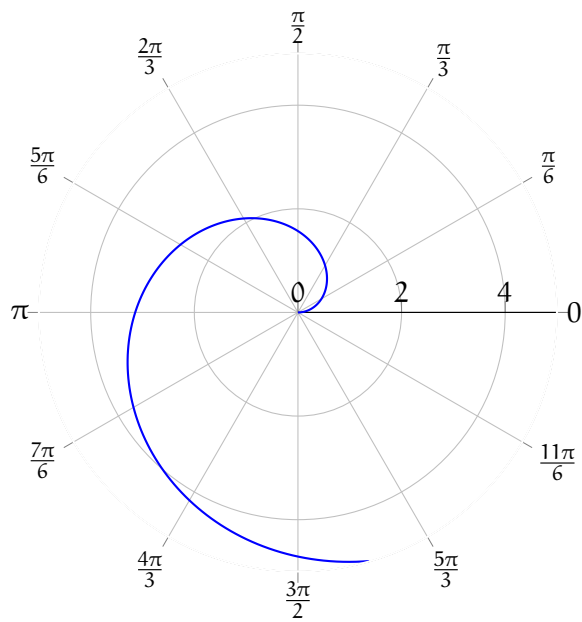
3.  $r = 2 \cos \frac{\theta}{2}$



4.  $r = -4 \sin \theta$



5.  $r = \theta$  (The spiral continues, but is beyond the boundary of the graph)







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